Highlights

Multi-Task Particle Swarm Optimization with Knowledge-Guided Elite Refinement and Two-Phase Weight Scheduling for High-Dimensional Feature Selection

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- Closed-loop knowledge-guided swarm learning framework for high-dimensional feature selection.
- MI initialization with adaptive re-initialization restores diversity and prevents search entrapment.
- Knowledge-guided elite refinement accumulates feature-importance signals with exponential decay.
- Two-phase weight scheduling temporally decouples feature compression from accuracy refinement.
- KDS-EL achieves 95.74% accuracy with 87.8 selected features across 20 benchmark datasets.

Multi-Task Particle Swarm Optimization with Knowledge-Guided Elite Refinement and Two-Phase Weight Scheduling for High-Dimensional Feature Selection

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ABSTRACT

Feature selection in high-dimensional data remains a central challenge for evolutionary computation. Particle swarm optimization suffers from three structural deficiencies: random initialization provides insufficient guidance in sparse high-dimensional spaces, uniform feature perturbation fails to exploit cross-iteration feature-importance signals, and static objective preferences cannot adapt to different evolutionary phases. To address these limitations, we propose KDS-EL (Knowledge-Guided Dual-Phase Swarm Evolutionary Learning), a knowledge-guided evolutionary framework with a closed-loop feature-importance learning mechanism. First, to overcome inefficient initialization, we propose mutual-information-guided initialization with adaptive re-initialization, injecting prior knowledge and recovering population diversity when stagnation occurs. Second, to exploit discarded cross-iteration feature-importance signals, we propose a knowledge-guided elite refinement strategy that accumulates feature-importance knowledge and steers elite solutions toward historically effective features. Third, to align objective weights with search phases, we propose two-phase weight scheduling that emphasizes early dimensionality reduction and late-stage accuracy refinement. On 20 high-dimensional benchmarks, KDS-EL achieves the best classification performance among all competitors, with a mean accuracy of 0.9574 and only 87.8 selected features, outperforming the strongest single- and multi-objective competitors by +3.97 and +4.42 percentage points; Wilcoxon signed-rank tests with Holm–Bonferroni correction confirm statistically significant improvements over all 13 competitors.

1. Introduction

High-dimensional data have emerged as a fundamental challenge in classification problems across diverse domains, including genomics, text mining, and computer vision. These datasets typically contain hundreds or even tens of thousands of features, while only a limited number of labeled samples are available, resulting in the well-known $p \gg n$ scenario. Under such conditions, conventional learning algorithms often suffer from severe overfitting, increased computational complexity, and reduced interpretability due to the curse of dimensionality [1, 2]. Feature selection (FS) has therefore become an essential preprocessing technique that aims to identify a compact and informative subset of features while preserving classification performance. However, selecting an optimal feature subset remains highly challenging because the search space grows exponentially with the number of features D, and the underlying subset-selection problem is NP-hard [3]. Moreover, individual feature relevance measures are insufficient to characterize complex feature interactions and dependencies, which are often critical for achieving robust classification performance.

FS methods are generally categorized into filter, wrapper, and embedded approaches [1, 4]. Filter methods rank features according to statistical criteria and offer high computational efficiency, but they usually evaluate features independently of the classifier and may overlook feature interactions relevant to classification. Embedded methods incorporate feature selection into the learning process, providing a compromise between selection efficiency and predictive performance; however, their effectiveness is often dependent on the characteristics of specific learning models. Wrapper methods directly optimize feature subsets according to classifier performance and can achieve competitive classification results, but the repeated model evaluations make them computationally expensive, especially in high-dimensional search spaces. To address the combinatorial complexity of wrapper-based FS, evolutionary computation has been extensively investigated as a promising optimization paradigm [3, 5–7]. By maintaining a population of candidate solutions, evolutionary algorithms can perform global exploration of feature subsets without requiring gradient information. Among them, particle swarm optimization (PSO), genetic algorithms, and differential evolution have demonstrated promising results on high-dimensional FS tasks.

PSO has become one of the most widely adopted evolutionary algorithms for FS due to its simple structure, efficient information-sharing mechanism, and fast convergence. Nevertheless, conventional PSO-based FS methods still encounter three structural limitations when dealing with sparse and high-dimensional search spaces. First, random initialization distributes particles without prior guidance throughout the binary search space, whereas effective feature subsets usually occupy only a small region with limited active dimensions. Consequently, a considerable number of fitness evaluations may be wasted on redundant and suboptimal solutions before the swarm identifies promising search regions. Second, conventional uniform feature perturbation strategies treat all dimensions equally and neglect the feature-level improvement information accumulated during previous evaluations, preventing the search from exploiting historically informative features. Third, existing PSO-based FS methods typically employ fixed objective weights to balance classification error and feature subset size during the entire optimization process. However, the desired trade-off between feature reduction and accuracy refinement varies across search stages: aggressive sparsity enhancement is preferred in early exploration, whereas accurate decision boundary refinement becomes more important after a compact subset has been obtained.

Recent evolutionary FS studies have attempted to improve search performance through enhanced population initialization, adaptive evolutionary operators, and refined fitness evaluation strategies [8–11]. Although these approaches have achieved promising results, they usually focus on isolated components of the optimization process and do not jointly address inefficient initialization, insufficient exploitation of accumulated feature knowledge, and inflexible objective weighting. Recently, multi-task PSO has emerged as an effective paradigm for high-dimensional FS with fixed-length binary representations. For instance, MEL [12] adopts cooperative subpopulations and a fixed-length representation to efficiently handle feature spaces containing tens of thousands of dimensions. Despite its effectiveness, existing multi-task PSO-based FS frameworks still leave several challenges unresolved: the initial population is generally generated without sufficient feature-level prior knowledge, useful feature-importance information discovered during evolution is not explicitly retained and exploited, and the balance between classification performance and feature sparsity remains unchanged across different search stages.

To address these limitations, this paper proposes Knowledge-Guided Dual-Phase Swarm Evolutionary Learning (KDS-EL), a closed-loop framework that synergistically integrates domain prior injection, cross-iteration knowledge accumulation, and dynamic objective scheduling into a cooperative multi-task PSO backbone.

The main contributions of this paper are summarized as follows:

1. **MI-Guided Initialization with Adaptive Re-injection**: A hybrid seeding strategy that injects continuous feature-relevance priors into population construction to eliminate early-stage search waste, coupled with an adaptive re-injection mechanism that restores swarm diversity upon evolutionary stagnation.
2. **Knowledge-Guided Elite Refinement (KER)**: A targeted elite perturbation operator that dynamically harvests cross-iteration feature-importance feedback with exponential memory decay, directing elite solutions toward historically effective features within a strictly bounded evaluation overhead.

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Table 1: Taxonomic and methodological comparison between KDS-EL and representative evolutionary feature selection algorithms.

		Knowledge Engineering			Search Control		
Algorithm	Paradigm	MI Prior Seeding	Online Memory	Memory Decay	Elite Refinement	Dynamic Weighting	Complexity
BPSO [7]	SO-SI	×	×	×	×	×	High
FWPSO [8]	SO-SI	○	×	×	×	×	Moderate
SaWDE [9]	SO-DE	×	×	×	×	○	High
W-MOPSO-FS [13]	MO-SI	×	×	×	×	×	High
NNS-DFPG-CSA [14]	SO-CSA	○	×	×	×	×	Moderate
MI-MPODE [15]	SO-DE	✓	×	×	×	×	Moderate
EMSWOA [16]	MS-WOA	×	×	×	○	×	High
MEL [12]	MT-SI	×	✓	×	×	×	High
KDS-EL (Proposed)	**MT-SI**	✓	✓	✓	✓	✓	**Low**

Note: ✓: Fully supported; ○: Partially supported (e.g., hard pre-filtering truncation, static velocity biasing, or adaptive scalar mutation); ×: Absent / not supported.
Paradigms: SO: Single-Objective; MO: Multi-Objective; MS: Multi-Swarm; MT: Multi-Task; SI: Swarm Intelligence; DE: Differential Evolution; CSA: Crow Search Algorithm; WOA: Whale Optimization Algorithm.

3. **Two-Phase Weight Scheduling (TPS)**: A stage-aware objective scheduling mechanism that temporally decouples early aggressive feature pruning from late-stage accuracy refinement, overcoming the rigidity of static weighting.
4. **Comprehensive Empirical Validation**: Extensive evaluations across 20 high-dimensional benchmark datasets (spanning 500 to 54,675 features) against 13 state-of-the-art competitors, rigorously validated through non-parametric statistical testing, component contribution analysis, and parametric sensitivity investigations.

Extensive experiments demonstrate that KDS-EL achieves the best classification performance, obtaining an average classification accuracy of 0.9574 with only 87.8 selected features. Compared with the strongest single-objective and multi-objective competitors, KDS-EL improves accuracy by 3.97 and 4.42 percentage points, respectively, while achieving a 3.4× speedup over the MEL baseline (reducing average runtime from 175.0 s to 51.4 s). Furthermore, Wilcoxon signed-rank tests with Holm–Bonferroni correction confirm that the improvements are statistically significant over all 13 competitors.

The remainder of this paper is organized as follows. Section 2 reviews feature selection methods and evolutionary FS algorithms. Section 3 introduces the background knowledge, including PSO, mutual-information-based feature relevance estimation, and the MEL framework. Section 4 presents the proposed KDS-EL framework and describes its three mechanisms in detail. Section 5 reports the experimental configuration and results, including comparative experiments, component contribution analysis, and parameter sensitivity analysis. Finally, Section 6 concludes this paper and discusses future research directions.

2. Related Work

This section provides a systematic survey of evolutionary feature selection paradigms, multi-objective versus single-objective formulations in high-dimensional spaces, evolutionary multi-task optimization, and knowledge-guided meta-heuristics, establishing the theoretical demarcation and unique positioning of KDS-EL.

2.1. Evolutionary Swarm Intelligence for Feature Selection

Feature selection is a combinatorial NP-hard optimization problem that aims to identify an optimal subset of $d \ll D$ relevant features, maximizing predictive accuracy while eliminating redundant and uninformative dimensions [1, 2]. Traditional feature selection techniques are categorized into three classical paradigms:

- **Filter methods**: Filter techniques evaluate feature subsets based on intrinsic statistical properties, such as mutual information (mRMR [17]), correlation (FCBF [18]), or graph-theoretic centrality [19]. Although computationally efficient ($O(D \log D)$), filter methods evaluate features independently or pairwise, fundamentally ignoring complex multivariate interactions and resulting in suboptimal accuracy when applied to non-linear classification landscapes.

- **Wrapper methods**: Wrapper approaches evaluate candidate subsets using the direct predictive feedback of a machine learning classifier [4]. While wrappers capture intricate feature dependencies and deliver superior classification performance, exhaustive evaluation is intractable in high-dimensional spaces (2^D combinatorial states), necessitating stochastic global search heuristics.

- **Embedded methods**: Embedded methods incorporate feature selection directly into model training via regularization, such as LASSO/Bolasso [20] and tree-based ensemble importance (FS-GBDT [21]). However, embedded rankings are tightly coupled with specific classifier inductive biases and frequently struggle to generalize across diverse classification architectures.

To harness the predictive fidelity of wrappers while mitigating combinatorial computational bottlenecks, Swarm Intelligence (SI) and Evolutionary Computation (EC) have emerged as the dominant paradigms for feature selection [3, 22]. Classical meta-heuristics, including Binary Particle Swarm Optimization (BPSO) [7], Genetic Algorithms (GA) [23], Ant Colony Optimization (ACO) [24], and Artificial Bee Colony (ABC) [25], navigate high-dimensional spaces via population-based stochastic exploration.

To combat the curse of dimensionality in small-sample regimes ($p \gg n$), numerous specialized single-objective evolutionary algorithms have been developed: self-adaptive swarm mechanisms [26], progressively refined search trajectories [27], self-adaptive differential evolution with adaptive weighting (SaWDE) [9], multi-layer perceptron-guided PSO (IGPSO) [10], multi-criterion scoring PSO (PSO-CSM) [11], reinforcement-learning-assisted swarm control [28], and divide-and-conquer subspace decomposition [29]. While these methods enhance local search operators, they universally treat feature perturbation as a memoryless stochastic trial, discarding the accumulated evolutionary feedback across generations.

2.2. Multi-Objective versus Single-Objective Formulations at Scale

Feature selection naturally involves two conflicting objectives: minimizing classification error and minimizing the number of selected features. Multi-Objective Feature Selection (MOFS) algorithms model this trade-off by searching for a Pareto front of non-dominated solutions. Pioneered by Xue et al. [13], diverse multi-objective evolutionary algorithms have been adapted to feature selection, including NSGA-II-FS [30], MOEA/D-FS [31], MOPSO-FS [32], and hybrid filter-wrapper multi-objective architectures [33, 34].

Despite their theoretical appeal, MOFS methods encounter severe operational limitations when scaled to ultra-high-dimensional biomedical data ($D > 10^3$):

1. **Combinatorial archive and sorting overhead**: Maintaining an external non-dominated archive and executing Pareto ranking over dense high-dimensional vectors incurs substantial computational overhead, often exceeding 110–140 seconds in wall-clock time, as analyzed in Section 5.7.
2. **Pareto front dilution in ultra-high dimensions**: In extreme $p \gg n$ regimes, non-dominated sorting tends to produce large, diffused Pareto fronts populated by non-sparse candidate subsets (often retaining 2,000–5,000 active features), failing to deliver the ultra-sparse compact biomarker signatures with fewer than 100 features required for clinical interpretability.

To circumvent Pareto archive bloat and achieve extreme subset sparsity within tight computational budgets, weighted single-objective scalarization remains the preferred formulation in high-dimensional biomarker discovery [12,

16]. However, conventional weighted frameworks enforce fixed static weights throughout search (such as $\mathbf{w} = [0.90, 0.10]$), which fails to adapt to the distinct evolutionary phases of feature selection. KDS-EL overcomes this limitation by introducing a dynamic Two-Phase Weight Scheduling strategy that aggressively eliminates noise features early and fine-tunes classification accuracy late.

2.3. Evolutionary Multi-Task Optimization in Feature Selection

Evolutionary Multi-Task Optimization (EMTO) aims to solve multiple distinct optimization tasks simultaneously by facilitating inter-task knowledge transfer through a unified chromosomal representation [35]. In feature selection, EMTO frameworks partition the global population into collaborative subpopulations to explore complementary search subspaces. Prominent multi-task feature selection frameworks include minion-assisted multi-tasking [36], evolutionary multi-task PSO for hyperspectral and gene expression data [37, 38], multi-task bee colony band selection [39], and large-scale multi-objective multi-tasking [40].

Most notably, Zhang et al. [12] introduced Multi-Task Evolutionary Learning (MEL), which establishes a dual-subpopulation PSO framework. In MEL, Subpopulation 1 performs continuous swarm exploration guided by inter-task attractors, while Subpopulation 2 conducts discrete feature sampling based on a shared feature-importance matrix. While MEL successfully demonstrates the power of cross-task knowledge transfer, its baseline implementation suffers from three critical bottlenecks: (i) random uniform initialization that causes unguided early-stage exploration in sparse spaces, (ii) unconstrained feature importance accumulation without memory discounting, and (iii) rigid static objective weights that delay early dimensionality reduction.

2.4. Knowledge-Guided Meta-Heuristics and Information-Theoretic Hybridization

Integrating domain-specific prior knowledge into meta-heuristics has attracted surging attention as a means to escape the blindness of black-box search [14, 15, 41]. In high-dimensional feature selection, mutual information (MI) serves as the primary information-theoretic metric to quantify non-linear feature relevance [42].

Existing knowledge-guided evolutionary algorithms typically adopt one-shot offline injection:

- **Static filter pre-filtering**: Algorithms such as MI-MPODE [15] and NNS-DFPG-CSA [14] compute MI rankings offline and initialize the population or restrict the search space to the top-k ranked features.
- **Static velocity biasing**: Frameworks such as FWPSO [8] incorporate static feature-weight vectors to bias particle velocity equations throughout evolution.

While beneficial, one-shot knowledge injection lacks online adaptability: once injected, the static prior is never updated or refined by the empirical trial-and-error feedback generated during subsequent wrapper evaluations. Conversely, learning-based surrogates (such as neural networks [10] or Gaussian processes [43]) introduce severe surrogate training overhead and risk model misspecification in extreme $p \gg n$ regimes.

2.5. Unique Positioning and Structural Advancements of KDS-EL

To overcome the collective limitations identified in the existing literature, KDS-EL establishes a closed-loop, knowledge-guided evolutionary learning framework that unifies offline domain priors, online memory harvesting, and stage-aware objective scheduling:

1. **Soft filter-wrapper seeding vs. hard truncation**: Unlike conventional filter wrappers that permanently eliminate features via hard truncation, KDS-EL uses normalized MI scores as continuous Gaussian-perturbed seeds, preserving the full search space while ensuring an informative, sparse starting basin ($\|\mathbf{x}\|_0 \ll D/2$).
2. **Online cumulative knowledge distillation vs. memoryless perturbation**: In contrast to black-box swarm mutations, KDS-EL dynamically harvests evolutionary fitness feedback into a cumulative importance matrix W, regulated by an exponential forgetting factor $\gamma = 0.99$. The KER operator re-deploys this accumulated knowledge to execute targeted, directional elite flips within a strictly bounded computational budget with $k_e = 3 \ll N$.
3. **Dynamic stage-aware objective scheduling vs. static weights**: Unlike rigid static weighting ($\mathbf{w} = [0.90, 0.10]$), KDS-EL's Two-Phase Weight Scheduling temporally decouples coarse feature pruning in Phase 1 ($w_f = 0.30$) from fine accuracy refinement in Phase 2 ($w_a = 0.95$), achieving simultaneous gains in accuracy, feature compactness, and runtime efficiency.

Table 1 summarizes the structural comparison between KDS-EL and representative state-of-the-art evolutionary feature selection frameworks.

3. Preliminaries

This section establishes the theoretical foundations, mathematical notations, and baseline architectures that underpin our proposed framework. We first formulate classical continuous and binary particle swarm optimization, followed by the information-theoretic definition of mutual information for feature relevance estimation, and conclude with the dual-subpopulation multi-task evolutionary learning (MEL) baseline.

3.1. Particle Swarm Optimization and Binary Formulations

Particle Swarm Optimization (PSO), originally introduced by Kennedy and Eberhart [5], is a population-based stochastic optimization paradigm inspired by the collective foraging behaviors of bird flocks. A swarm consists of N particles navigating a D-dimensional search space $\mathbb{R}^D$. Each particle $i \in \{1, \ldots, N\}$ maintains a continuous position vector $\mathbf{p}_i^{(t)} = [p_{i,1}^{(t)}, \ldots, p_{i,D}^{(t)}]$ and a continuous velocity vector $\mathbf{v}_i^{(t)} = [v_{i,1}^{(t)}, \ldots, v_{i,D}^{(t)}]$ at iteration t. The historical trajectory is governed by its personal best position $\mathbf{p}_{\text{best},i}$ and the swarm's global best position $\mathbf{g}_{\text{best}}$.

The velocity and position updates are formulated as:

$$\mathbf{v}_i^{(t+1)} = \omega \mathbf{v}_i^{(t)} + c_1 r_1 (\mathbf{p}_{\text{best},i} - \mathbf{p}_i^{(t)}) + c_2 r_2 (\mathbf{g}_{\text{best}} - \mathbf{p}_i^{(t)}), \tag{1}$$

$$\mathbf{p}_i^{(t+1)} = \mathbf{p}_i^{(t)} + \mathbf{v}_i^{(t+1)}, \tag{2}$$

where ω denotes the inertia weight governing momentum retention [6], c_1 and c_2 are cognitive and social acceleration coefficients, and $r_1, r_2 \sim \mathcal{U}(0, 1)$ are independent uniform random stochastic factors.

To adapt PSO to combinatorial binary feature selection where solutions represent binary indicator masks $\mathbf{x}_i \in \{0, 1\}^D$ ($x_{i,j} = 1$ indicates feature j is selected), Kennedy and Eberhart [7] proposed the standard Binary PSO (BPSO). BPSO transforms continuous velocities into selection probabilities via a Sigmoid transfer function:

$$S(v_{i,j}^{(t+1)}) = \frac{1}{1 + \exp(-v_{i,j}^{(t+1)})}, \quad x_{i,j}^{(t+1)} = \begin{cases} 1, & \text{if rand}() < S(v_{i,j}^{(t+1)}), \\ 0, & \text{otherwise.} \end{cases} \tag{3}$$

While effective for low-dimensional problems, standard BPSO suffers severe exploration-exploitation dilemmas in ultra-high-dimensional spaces with $p \gg n$: continuous velocity updates provide no domain-grounded guidance toward sparse, informative biomarker subspaces. In KDS-EL, particles maintain continuous normalized position coordinates $\mathbf{p}_i \in [0, 1]^D$, mapped deterministically to binary feature masks via a strict thresholded Heaviside step function $x_{i,j} = \mathbb{I}(p_{i,j} \geq \theta)$ with $\theta = 0.60$, eliminating stochastic binarization variance.

3.2. Mutual Information for Information-Theoretic Feature Relevance

Mutual information (MI), rooted in Shannon's information theory [42], quantifies the mutual dependence and reduction in uncertainty between two random variables. For a feature variable f_j and the discrete class label $Y \in \{1, \ldots, C\}$, the mutual information $I(f_j; Y)$ is defined as:

$$I(f_j; Y) = H(Y) - H(Y \mid f_j) = \sum_{v \in \text{Val}(f_j)} \sum_{c=1}^{C} P(f_j = v, Y = c) \log_2 \frac{P(f_j = v, Y = c)}{P(f_j = v) P(Y = c)}, \tag{4}$$

where $H(Y)$ is the marginal Shannon entropy of the class labels and $H(Y \mid f_j)$ is the conditional entropy given feature f_j. High values of $I(f_j; Y)$ indicate strong discriminative relevance to classification targets.

Conventional filter feature selection methods [14, 17] evaluate $I(f_j; Y)$ across all D dimensions and execute a hard truncation:

$$\mathcal{S}_{\text{Filter}} = \{\text{IDX}(1), \text{IDX}(2), \ldots, \text{IDX}(\lfloor k \cdot D \rfloor)\}, \tag{5}$$

where IDX denotes the feature indices sorted in descending order of $I(f_j; Y)$ and $k \in (0, 1]$ represents the retention ratio, permanently discarding all features outside the top-k ranking. While computationally rapid ($O(D \log D)$), hard truncation fundamentally ignores multi-feature synergy: individual features with moderate MI scores that are collectively decisive when combined are permanently eliminated. Rather than using hard truncation, KDS-EL utilizes normalized MI scores as a soft prior to construct continuous initial population seeds (Section 4.2), injecting filter knowledge into the evolutionary wrapper without irreversibly shrinking the search space.

3.3. Multi-Task Evolutionary Learning Baseline (MEL)

Multi-Task Evolutionary Learning (MEL), recently proposed by Zhang et al. [12], introduces a cooperative multi-task architecture for high-dimensional feature selection. In MEL, a population of N particles is partitioned into two collaborative subpopulations, $\mathcal{P}_1$ and $\mathcal{P}_2$ ($|\mathcal{P}_1| = |\mathcal{P}_2| = N/2$), operating on dual complementary search tasks:

Subpopulation 1 ($\mathcal{P}_1$): Continuous swarm exploration. Particles update continuous velocities and positions using PSO dynamics, augmented with an inter-task knowledge transfer channel directing particles toward the historical best of Subpopulation 2 ($\mathbf{g}_{\text{best},2}$):

$$\mathbf{v}_{1,i}^{(t+1)} = \omega \mathbf{v}_{1,i}^{(t)} + c_1 r_1 (\mathbf{p}_{\text{best},1,i} - \mathbf{p}_{1,i}^{(t)}) + c_2 r_2 (\mathbf{g}_{\text{best},1} - \mathbf{p}_{1,i}^{(t)}) + c_3 r_3 (\mathbf{g}_{\text{best},2} - \mathbf{p}_{1,i}^{(t)}). \quad (6)$$

Subpopulation 2 ($\mathcal{P}_2$): Importance-guided discrete sampling. Rather than continuous mechanics, $\mathcal{P}_2$ samples binary feature masks directly based on a shared feature-importance matrix $W_{\text{MEL}} \in \mathbb{R}^D$.

In MEL, both subpopulations share a scalar fitness function combining classification error and feature subset ratio via static weights:

$$f_{\text{MEL}}(\mathbf{x}) = \alpha \cdot (1 - \text{Acc}(\mathbf{x})) + \beta \cdot \left(\frac{\|\mathbf{x}\|_0}{D} \right), \quad (7)$$

where $\text{Acc}(\mathbf{x})$ is the 5-fold cross-validation accuracy of a k-NN classifier ($k = 3$), and $\alpha = 0.90, \beta = 0.10$ are fixed constants [12].

While MEL demonstrates the utility of cross-task knowledge transfer, its reliance on uniform random initialization, unconstrained importance accumulation without memory discounting, and rigid static weighting creates substantial search inefficiencies in ultra-high-dimensional spaces. The proposed KDS-EL framework directly inherits this dual-subpopulation backbone while resolving these deficiencies through three synergistic mechanisms detailed in Section 4.

4. Proposed KDS-EL Algorithm

In this section, we present the comprehensive architectural design, mathematical formulation, and algorithmic execution of the proposed Knowledge-Guided Dual-Phase Swarm Evolutionary Learning (KDS-EL) algorithm. KDS-EL is designed specifically to overcome three fundamental structural deficiencies of conventional particle swarm optimization in ultra-high-dimensional feature selection: (i) the search inefficiency and wasted evaluation budget caused by uniform random initialization in sparse feature spaces, (ii) the loss of historical cross-iteration feature-importance signals due to memoryless random perturbation, and (iii) the inability of static objective weighting to adapt to the divergent requirements of coarse feature reduction versus fine-grained accuracy refinement.

4.1. Overall Algorithmic Architecture

The macroscopic architecture of KDS-EL is illustrated in Fig. 1. Built upon a cooperative multi-task dual-subpopulation backbone [12], KDS-EL divides a population of $N = 20$ binary particles into two collaborative subpopulations of equal size ($N_1 = N_2 = 10$). Subpopulation 1 ($\mathcal{P}_1$) executes standard continuous-space PSO velocity-position updates with an inter-task knowledge transfer channel directing particles toward the historical global best of Subpopulation 2. Subpopulation 2 ($\mathcal{P}_2$) performs probability-driven feature sampling guided by the population's accumulated feature-importance landscape.

The optimization process operates in a closed loop through four tightly coupled functional stages:

1. **Filter-level quality bootstrapping (offline)**: Mutual information (MI) between each feature and class labels is computed once prior to evolution, establishing a normalized baseline relevance vector $\mathbf{m} \in [0, 1]^D$.
2. **Knowledge-informed hybrid seeding**: An initial population of N particles is generated by blending Gaussian-perturbed MI seeds with uniform random particles at a ratio of MI_init_ratio = 0.50, followed by Heaviside step binarization with threshold $\theta = 0.60$.
3. **Dynamic two-phase evolutionary loop**: Across $T = 100$ iterations, the objective weights $\mathbf{w}(t) = [w_a(t), w_f(t)]$ dynamically transition from aggressive feature pruning ($\mathbf{w}_1 = [0.70, 0.30]$ for $t \le 0.7T$) to accuracy-oriented refinement ($\mathbf{w}_2 = [0.95, 0.05]$ for $t > 0.7T$).
4. **Accumulative knowledge harvesting and elite refinement**: Fitness improvements during evaluation are harvested into a continuous feature-importance matrix $W \in \mathbb{R}^D$, discounted by an exponential forgetting factor $\gamma = 0.99$. The top elite_k = 3 elite individuals are refined via the Knowledge-guided Elite Refinement (KER) operator and admitted via 1-to-1 in-place replacement. An adaptive re-initialization mechanism monitors stagnation and re-seeds the lower half of the population if global-best fitness stalls for stall_patience = 20 generations.

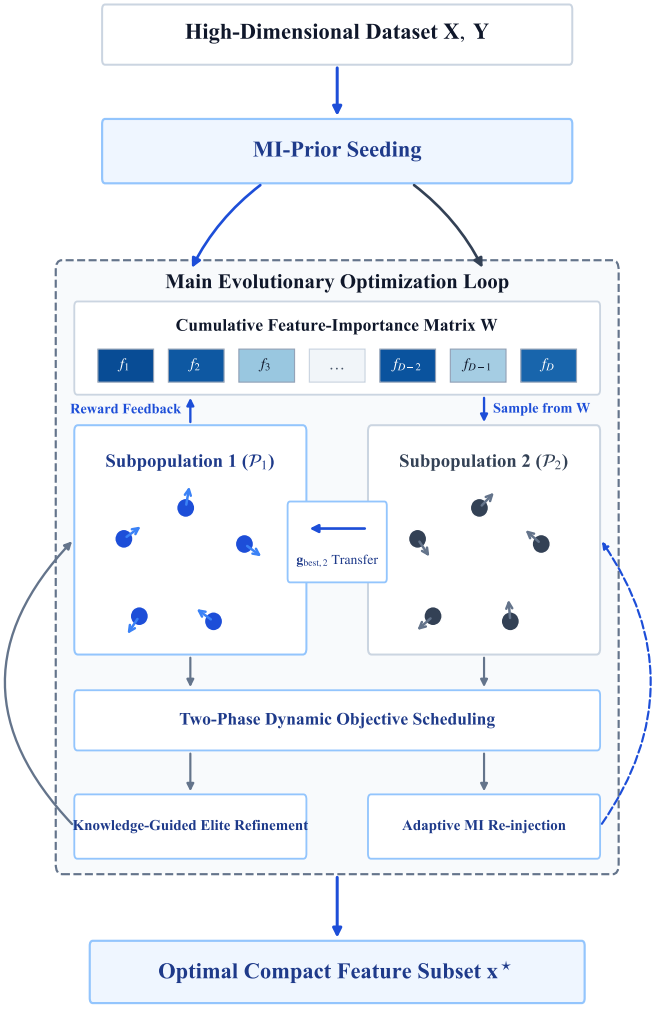


Figure 1: Overall framework of the proposed KDS-EL algorithm.

4.2. MI-Prior Seeding and Adaptive Re-injection Strategy

In ultra-high-dimensional feature selection ($D \gg 10^3$), standard uniform random initialization produces candidate solutions where approximately 50% of all features are active ($\|\mathbf{x}\|_0 \approx D/2$), forcing the swarm into dense, noisy search basins. To establish an informative initial search origin while preserving stochastic diversity, KDS-EL introduces a filter-wrapper hybrid seeding strategy.

Given the training dataset $\mathcal{D} = \{(\mathbf{f}_i, y_i)\}_{i=1}^{N_{\text{samp}}}$, the mutual information scores $\{I(f_j; Y)\}_{j=1}^D$ computed via (4) are normalized into a baseline prior relevance vector $\mathbf{m} \in [0, 1]^D$:

$$m_j = \frac{I(f_j; Y) - \min_{1 \le k \le D} I(f_k; Y)}{\max_{1 \le k \le D} I(f_k; Y) - \min_{1 \le k \le D} I(f_k; Y) + \epsilon}, \quad (8)$$

where $\epsilon = 10^{-12}$ prevents division by zero.

To prevent population homogenization while guiding search toward informative dimensions, the initial population is split into two halves based on $\rho = \text{MI_init_ratio} = 0.50$:

$$\mathbf{p}_i = \begin{cases} \text{clip}(\mathbf{m} + \boldsymbol{\varepsilon}_i,\ 0,\ 1), & \boldsymbol{\varepsilon}_i \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_D), \quad \text{for } 1 \le i \le \rho N, \\ \mathbf{u}_i, & \mathbf{u}_i \sim \mathcal{U}(0, 1)^D, \quad \text{for } \rho N < i \le N, \end{cases} \quad (9)$$

where $\sigma = 0.10$ represents the Gaussian noise perturbation scale and $\text{clip}(z, 0, 1) = \max(0, \min(1, z))$.

Continuous positions $\mathbf{p}_i = [p_{i,1}, \ldots, p_{i,D}]$ are mapped into binary feature selection vectors $\mathbf{x}_i \in \{0, 1\}^D$ using a strict Heaviside step binarization with threshold $\theta = 0.60$:

$$x_{i,j} = \begin{cases} 1, & \text{if } p_{i,j} \ge \theta, \\ 0, & \text{if } p_{i,j} < \theta. \end{cases} \quad (10)$$

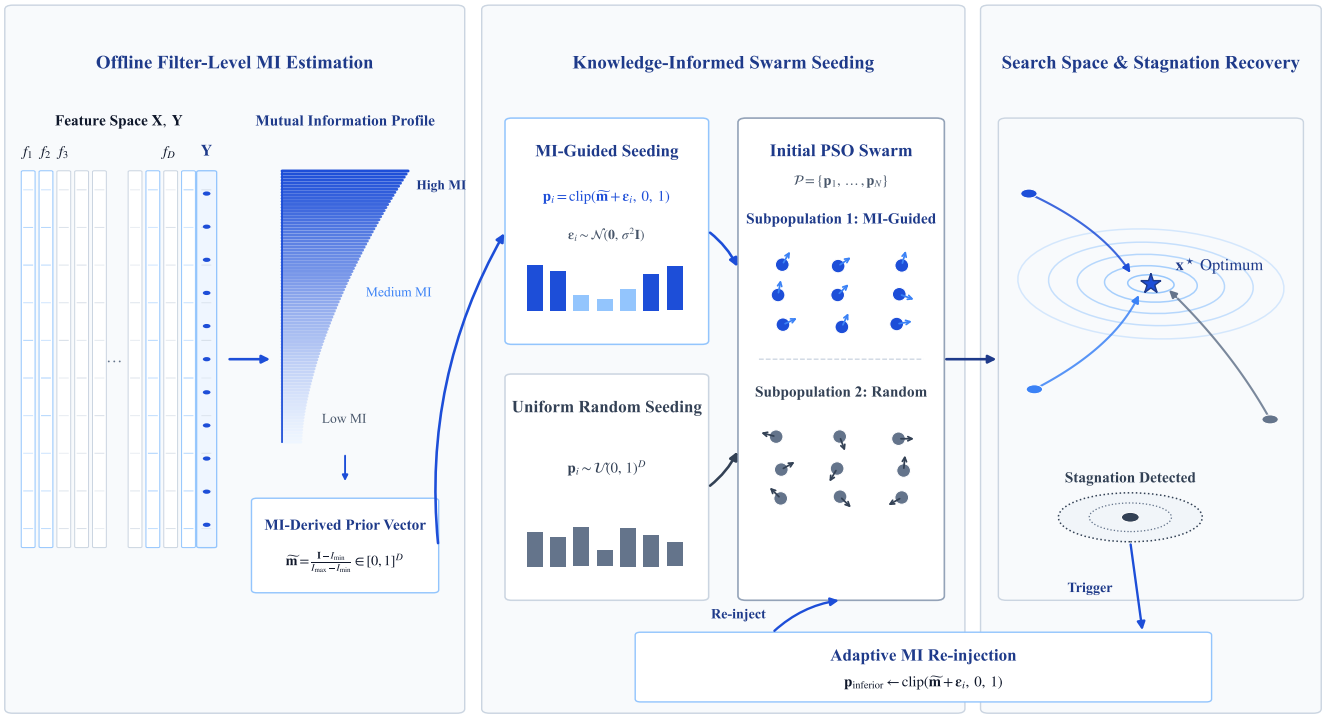


Figure 2: Schematic diagram of the MI-prior seeding and adaptive re-injection mechanism.

Because $\theta = 0.60 > 0.50$, only features possessing above-average mutual information are initialized as active, naturally yielding a sparse initial configuration ($\|\mathbf{x}_i\|_0 \approx 0.15D \sim 0.30D$) that accelerates early convergence, as formulated in Stage 2 of Section 4.6.

4.2.1. Adaptive Stagnation Re-initialization

High-dimensional multimodal search landscapes frequently cause swarm particles to become trapped in suboptimal local attractors. When the global-best fitness f_{gbest} fails to improve for $\tau = \text{stall_patience} = 20$ consecutive iterations, an adaptive re-initialization routine is executed. First, the population is sorted in ascending order of personal-best fitness f_i^{pbest}. The superior half ($\{i \mid \text{rank}(i) \leq N/2\}$) is preserved to safeguard historical elite trajectories, whereas the inferior half ($\{i \mid \text{rank}(i) > N/2\}$) is fully refreshed using the Gaussian-perturbed MI distribution:

$$\mathbf{p}_i \leftarrow \text{clip}(\mathbf{m} + \boldsymbol{\varepsilon}_i,\ 0,\ 1), \quad x_{i,j} \leftarrow \mathbb{I}(p_{i,j} \geq \theta). \tag{11}$$

The refreshed particles are then re-evaluated under the current objective weights, and the stagnation counter is reset to zero. This restores diversity within a high-potential subspace without completely discarding established elite knowledge.

4.3. Cumulative Feature-Importance Matrix with Exponential Memory Decay

To capture and retain the empirical trial-and-error feedback generated across generations, KDS-EL maintains a continuous cumulative feature-importance matrix $W \in \mathbb{R}^D$, initialized to $\mathbf{0}$.

At generation t, whenever a candidate particle $\mathbf{x}_i^{(t)}$ achieves a strict fitness improvement over its personal best ($f_i^{(t)} < f_i^{\text{pbest}}$), the differential feature subset $C_i^{(t)}$ containing features whose selection status changed is extracted:

$$C_i^{(t)} = \{j \in \{1, \ldots, D\} \mid x_{i,j}^{(t)} \neq p_{\text{best},i,j}\}. \tag{12}$$

Let $\delta_i^{(t)} = f_i^{\text{pbest}} - f_i^{(t)} > 0$ denote the fitness improvement. Each changed feature $j \in C_i^{(t)}$ receives a directional reward proportional to $\delta_i^{(t)}$:

$$W_j \leftarrow W_j + \delta_i^{(t)} \cdot (2x_{i,j}^{(t)} - 1), \quad \forall j \in C_i^{(t)}, \tag{13}$$

where $(2x_{i,j}^{(t)} - 1) = +1$ if feature j was newly activated and -1 if deactivated. Non-improving particles ($\delta_i^{(t)} \leq 0$) exert zero impact on W.

To prevent early-discovered features from dominating the matrix indefinitely and allow the algorithm to adapt as the search shifts toward fine-grained subspaces, an exponential memory decay is applied at the conclusion of each generation:

$$W \leftarrow \gamma \cdot W, \quad \text{where } \gamma = 0.99. \tag{14}$$

The decay factor ensures that unless a feature repeatedly contributes to fitness improvements, its score depreciates as γ^k, preventing score saturation while dynamically maintaining ranking sensitivity.

4.4. Knowledge-guided Elite Refinement (KER) Strategy

To exploit the historical evolutionary knowledge distilled in W, the Knowledge-guided Elite Refinement (KER) strategy performs targeted directional bit perturbations on top-performing solutions, selectively activating promising absent features and pruning uninformative active features.

At each generation, the top $\text{elite_k} = 3$ individuals (ranked by fitness) are selected. For each elite individual $\mathbf{x}_e \in \{0,1\}^D$, KER executes a probabilistic dual-branch operation (Fig. 3):

Branch 1 (Ergodic random exploration, $p_{\text{rand}} = 0.10$): To guarantee search ergodicity, $\lfloor \phi D \rfloor$ features are selected uniformly at random and inverted:

$$x'_{e,j} \leftarrow 1 - x_{e,j}, \quad \forall j \in \mathcal{S}_{\text{rand}}, \quad |\mathcal{S}_{\text{rand}}| = \lfloor \phi D \rfloor, \tag{15}$$

where $\phi = \text{flip_frac} = 0.01$ defines the refinement perturbation budget.

Branch 2 (Knowledge-guided directional refinement, $1 - p_{\text{rand}} = 0.90$): Each feature $j \in \{1, \ldots, D\}$ is assigned a state-dependent directional flip score σ_j:

$$\sigma_j = (1 - 2x_{e,j}) \cdot W_j = \begin{cases} +W_j, & \text{if } x_{e,j} = 0 \quad \text{(activate absent feature)}, \\ -W_j, & \text{if } x_{e,j} = 1 \quad \text{(deactivate present feature)}. \end{cases} \tag{16}$$

We identify the index set $\mathcal{S}_{\text{know}}$ containing the $\lfloor \phi D \rfloor$ features possessing the highest flip scores σ_j:

$$\mathcal{S}_{\text{know}} = \arg\max_{|\mathcal{S}| = \lfloor \phi D \rfloor} \sum_{j \in \mathcal{S}} \sigma_j, \quad \text{where } \mathcal{S} \subseteq \{1, \ldots, D\}, \tag{17}$$

and invert the selected features: $x'_{e,j} \leftarrow 1 - x_{e,j}, \forall j \in \mathcal{S}_{\text{know}}$.

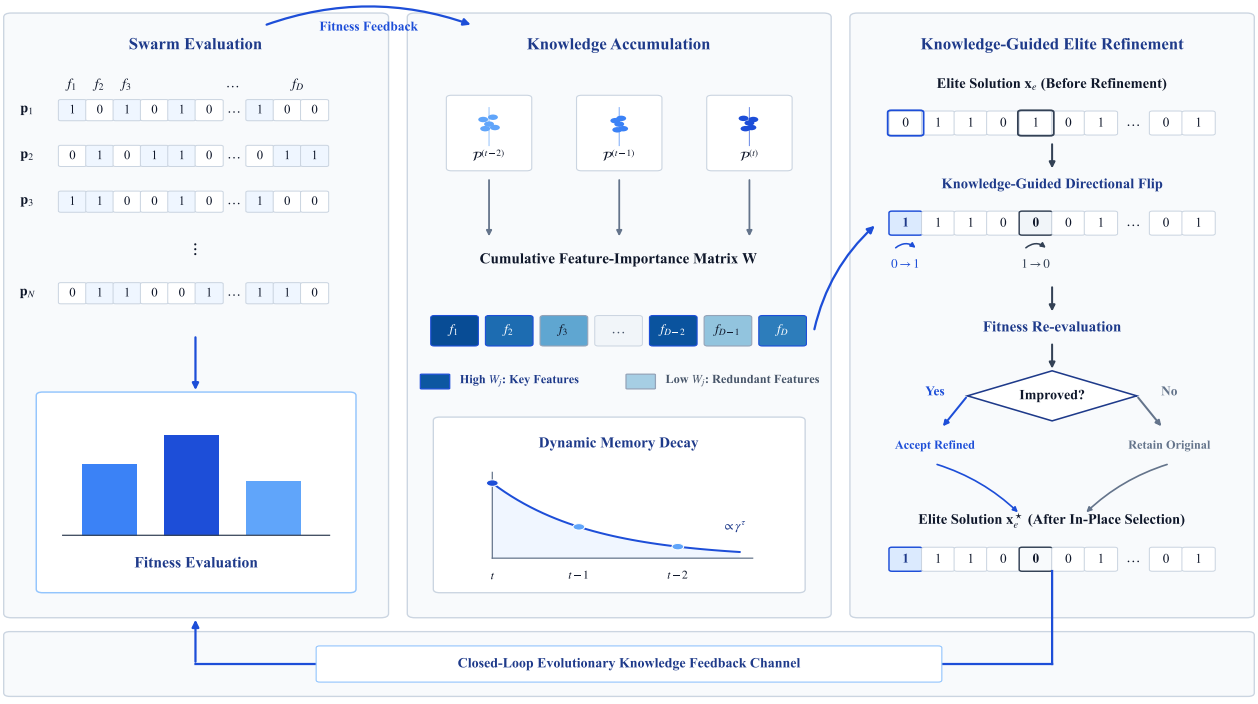


Figure 3: Flowchart of the Knowledge-guided Elite Refinement (KER) strategy.

4.4.1. 1-to-1 In-Place Replacement and Bounded Evaluation Budget

The refined mutant $\mathbf{x}'_e$ is evaluated in a closed loop, and admitted via greedy 1-to-1 in-place replacement:

$$\mathbf{x}_e^\star = \begin{cases} \mathbf{x}'_e, & \text{if } f(\mathbf{x}'_e, \mathbf{w}(t)) < f(\mathbf{x}_e, \mathbf{w}(t)), \\ \mathbf{x}_e, & \text{otherwise.} \end{cases} \tag{18}$$

If improved, $\mathbf{x}'_e$ immediately overwrites $\mathbf{x}_e$ in the population and updates the global best if $f(\mathbf{x}'_e) < f_{\text{gbest}}$. Because elite_k = 3 is fixed, KER executes within an exact bounded overhead of $k_e = 3$ evaluations per generation. The detailed execution of KER is formalized in Algorithm 1.

Algorithm 1 Knowledge-guided Elite Refinement (KER) strategy

Require: Elite candidate solution $\mathbf{x}_e \in \{0,1\}^D$, cumulative importance matrix $W \in \mathbb{R}^D$, perturbation budget $\phi \in (0,1)$, exploration probability $p_{\text{rand}} \in [0,1]$, dynamic objective weights $\mathbf{w}(t)$
Ensure: Refined elite candidate $\mathbf{x}_e^\star \in \{0,1\}^D$

$k_{\text{flip}} \leftarrow \max(1, \lfloor \phi \cdot D \rfloor)$ ▷ Determine perturbation budget
Sample random variable $r \sim \mathcal{U}(0,1)$
if $r < p_{\text{rand}}$ **then** ▷ Branch 1: Ergodic random exploration
  Sample random feature subset $\mathcal{S}_{\text{rand}} \subset \{1,\dots,D\}$ with $|\mathcal{S}_{\text{rand}}| = k_{\text{flip}}$
  $\mathbf{x}'_e \leftarrow \mathbf{x}_e$; $x'_{e,j} \leftarrow 1 - x_{e,j}$, $\forall j \in \mathcal{S}_{\text{rand}}$
else ▷ Branch 2: Knowledge-guided directional refinement
  for $j = 1$ **to** D **do**
    $\sigma_j \leftarrow (1 - 2x_{e,j}) \cdot W_j$ ▷ Compute directional flip score
  end for
  Identify subset $\mathcal{S}_{\text{know}}$ of size k_{flip} with top σ_j scores via (17)
  $\mathbf{x}'_e \leftarrow \mathbf{x}_e$; $x'_{e,j} \leftarrow 1 - x_{e,j}$, $\forall j \in \mathcal{S}_{\text{know}}$
end if
Evaluate offspring fitness $f' \leftarrow f(\mathbf{x}'_e, \mathbf{w}(t))$ using (20)
if $f' < f(\mathbf{x}_e, \mathbf{w}(t))$ **then**
  $\mathbf{x}_e^\star \leftarrow \mathbf{x}'_e$ ▷ Greedy 1-to-1 in-place replacement
else
  $\mathbf{x}_e^\star \leftarrow \mathbf{x}_e$ ▷ Retain original elite
end if
return $\mathbf{x}_e^\star$

4.5. Two-Phase Dynamic Objective Weight Scheduling Strategy

To reconcile the temporal trade-off between coarse feature reduction (early) and fine accuracy refinement (late), KDS-EL introduces the Two-Phase Weight Scheduling (TPS) strategy. The total generational budget $T = 100$ is partitioned at split point $\alpha = \text{PhaseSplit} = 0.70$. The dynamic dual-objective weight vector $\mathbf{w}(t) = [w_a(t), w_f(t)]$ is governed by:

$$\mathbf{w}(t) = \begin{cases} [0.70,\ 0.30], & \text{for } 1 \le t \le \lfloor \alpha T \rfloor \quad \text{(Phase 1: Aggressive Pruning)}, \\ [0.95,\ 0.05], & \text{for } \lfloor \alpha T \rfloor < t \le T \quad \text{(Phase 2: Accuracy Refinement)}. \end{cases} \tag{19}$$

The unified scalar fitness $f(\mathbf{x}, \mathbf{w}(t))$ of a binary feature subset $\mathbf{x} \in \{0,1\}^D$ at generation t is formulated as:

$$f(\mathbf{x}, \mathbf{w}(t)) = w_a(t) \cdot (1 - \text{Acc}(\mathbf{x})) + w_f(t) \cdot \left(\frac{\|\mathbf{x}\|_0}{D} \right), \tag{20}$$

where $\text{Acc}(\mathbf{x})$ denotes 5-fold cross-validation classification accuracy evaluated by a k-NN classifier ($k = 3$), and $\|\mathbf{x}\|_0/D$ represents the normalized feature selection ratio (FSR).

In Phase 1 ($t \le 0.7T$), the heavy feature-ratio penalty ($w_f = 0.30$) drives rapid elimination of non-informative dimensions. In Phase 2 ($t > 0.7T$), the weight switches to prioritize classification accuracy ($w_a = 0.95$), enabling fine-tuning of delicate biomarker interactions over the surviving compact subspace.

4.6. Integrated KDS-EL Main Algorithm

Algorithm 2 consolidates the holistic execution workflow of KDS-EL, integrating filter-based MI ranking, hybrid seeding, the dual-subpopulation evolutionary loop, cumulative importance matrix updating with exponential forgetting, and knowledge-guided elite refinement into a unified framework.

4.7. Asymptotic Computational and Space Complexity Analysis

To establish the computational efficiency of KDS-EL, we conduct a rigorous asymptotic complexity analysis covering all algorithmic components.

4.7.1. Time Complexity Decomposition

The execution time of KDS-EL is decomposed into two distinct phases:

Offline preprocessing phase (mutual information calculation): Computing the mutual information $I(f_j; Y)$ between each feature f_j and class label Y requires $O(N_{\text{samp}})$ operations per feature, where N_{samp} is the number of training samples. For all D features, computing and sorting the baseline MI vector incurs:

$$T_{\text{pre}} = O(D \cdot N_{\text{samp}} + D \log D). \tag{21}$$

Algorithm 2 Pseudo-code of the proposed KDS-EL algorithm

Require: Training dataset $\mathcal{D} = (X, Y)$ with D features, population size N, maximum iterations T, binarization threshold θ, MI seeding ratio ρ, memory decay factor γ, elite pool size k_e, perturbation fraction ϕ, random branch probability p_{rand}, phase split ratio α, stagnation threshold τ

Ensure: Optimal compact feature subset $\mathbf{x}^\star \in \{0,1\}^D$ and optimal fitness f_{gbest}

// Stage 1: Filter-level MI relevance extraction

Compute continuous mutual information $I(f_j; Y)$ for all features $j \in \{1, \ldots, D\}$ via (4)

Normalize MI scores to construct baseline prior relevance vector $\mathbf{m} \in [0,1]^D$ via (8)

// Stage 2: Knowledge-informed hybrid population seeding

for $i = 1$ **to** N **do**

  if $i \le \lfloor \rho N \rfloor$ **then**

    Sample Gaussian perturbation $\boldsymbol{\varepsilon}_i \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_D)$ and set continuous position $\mathbf{p}_i \leftarrow \text{clip}(\mathbf{m} + \boldsymbol{\varepsilon}_i, 0, 1)$

  else

    Sample continuous position uniformly: $\mathbf{p}_i \sim \mathcal{U}(0,1)^D$

  end if

  Binarize candidate solution: $x_{i,j} \leftarrow \mathbb{I}(p_{i,j} \ge \theta), \quad \forall j \in \{1, \ldots, D\}$

  Initialize velocity $\mathbf{v}_i \leftarrow \mathbf{0} \in \mathbb{R}^D$

end for

Initialize cumulative feature-importance matrix $W \leftarrow \mathbf{0} \in \mathbb{R}^D$ and stagnation counter stall_count $\leftarrow 0$

Evaluate initial population fitness $\{f_i\}_{i=1}^N$ under $\mathbf{w}(1) = [0.70, 0.30]$ via (20)

Initialize personal bests $\mathbf{p}_{\text{best},i} \leftarrow \mathbf{x}_i, \; f_i^{\text{pbest}} \leftarrow f_i, \; \forall i \in \{1, \ldots, N\}$

Identify initial global best $\mathbf{g}_{\text{best}} \leftarrow \arg\min_{\mathbf{p}_{\text{best},i}} f_i^{\text{pbest}}$ with fitness $f_{\text{gbest}} \leftarrow \min_i f_i^{\text{pbest}}$

// Stage 3: Dynamic two-phase evolutionary optimization loop

for $t = 1$ **to** T **do**

  Update dynamic objective weight vector $\mathbf{w}(t) \leftarrow \begin{cases} [0.70,\ 0.30], & \text{if } t \le \lfloor \alpha T \rfloor \quad \text{(Phase 1: Feature Pruning)} \\ [0.95,\ 0.05], & \text{if } t > \lfloor \alpha T \rfloor \quad \text{(Phase 2: Accuracy Refinement)} \end{cases}$

  // Subpopulation 1 ($\mathcal{P}_1$): Continuous PSO search with inter-task transfer

  for $i = 1$ **to** $N/2$ **do**

    $\mathbf{v}_{1,i} \leftarrow \omega \mathbf{v}_{1,i} + c_1 r_1 (\mathbf{p}_{\text{best},1,i} - \mathbf{p}_{1,i}) + c_2 r_2 (\mathbf{g}_{\text{best},1} - \mathbf{p}_{1,i}) + c_3 r_3 (\mathbf{g}_{\text{best},2} - \mathbf{p}_{1,i})$ ▷ Inter-task transfer from $\mathcal{P}_2$

    $\mathbf{p}_{1,i} \leftarrow \mathbf{p}_{1,i} + \mathbf{v}_{1,i}; \quad x_{1,i,j} \leftarrow \mathbb{I}(p_{1,i,j} \ge \theta), \quad \forall j \in \{1, \ldots, D\}$

  end for

  // Subpopulation 2 ($\mathcal{P}_2$): Probability-driven feature sampling guided by W

  for $i = N/2 + 1$ **to** N **do**

    Compute feature activation probabilities: $P_j \leftarrow \frac{1}{1+\exp(-W_j)}, \quad \forall j \in \{1, \ldots, D\}$

    Sample discrete candidate mask $\mathbf{x}_{2,i}$ based on $\mathbf{P}$ blended with personal best $\mathbf{p}_{\text{best},2,i}$

  end for

  // Fitness wrapper evaluation & differential knowledge harvesting

  for $i = 1$ **to** N **do**

    Evaluate candidate fitness $f_i \leftarrow f(\mathbf{x}_i, \mathbf{w}(t))$ using 5-fold cross-validation k-NN classifier via (20)

    if $f_i < f_i^{\text{pbest}}$ **then**

      Compute fitness improvement magnitude $\delta_i \leftarrow f_i^{\text{pbest}} - f_i$

      for each $j \in \{1, \ldots, D\}$ **such that** $x_{i,j} \ne p_{\text{best},i,j}$ **do**

        $W_j \leftarrow W_j + \delta_i \cdot (2x_{i,j} - 1)$ ▷ Reward activated (+1) or eliminated (-1) features

      end for

      Update personal best: $\mathbf{p}_{\text{best},i} \leftarrow \mathbf{x}_i, \quad f_i^{\text{pbest}} \leftarrow f_i$

    end if

  end for

  $W \leftarrow \gamma \cdot W$ ▷ Exponential memory decay via (14)

  // Knowledge-guided elite refinement

  Identify top k_e elite candidates $\mathcal{E} \subset \{1, \ldots, N\}$ ranked by f_i^{pbest}

  for each $e \in \mathcal{E}$ **do**

    $\mathbf{p}_{\text{best},e} \leftarrow \text{KER}(\mathbf{p}_{\text{best},e}, W, \phi, p_{\text{rand}}, \mathbf{w}(t))$ ▷ Call Algorithm 1

  end for

  // Global best tracking & adaptive stagnation recovery

  Update global best $\mathbf{g}_{\text{best}} \leftarrow \arg\min_{\mathbf{p}_{\text{best},i}} f_i^{\text{pbest}}, \; f_{\text{gbest_curr}} \leftarrow \min_i f_i^{\text{pbest}}$

  if $f_{\text{gbest_curr}} < f_{\text{gbest}}$ **then**

    $f_{\text{gbest}} \leftarrow f_{\text{gbest_curr}}, \quad$ stall_count $\leftarrow 0$

  else

    stall_count $\leftarrow$ stall_count + 1

  end if

  if stall_count $\ge \tau$ **then** ▷ Stagnation triggered: re-seed inferior half of swarm

    Sort population indices π by f_i^{pbest}

    for $k = \lfloor N/2 \rfloor + 1$ **to** N **do**

      Sample $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_D), \quad \mathbf{p}_{\pi(k)} \leftarrow \text{clip}(\mathbf{m} + \boldsymbol{\varepsilon}, 0, 1), \quad \mathbf{x}_{\pi(k)} \leftarrow \mathbb{I}(\mathbf{p}_{\pi(k)} \ge \theta)$

      $\mathbf{p}_{\text{best},\pi(k)} \leftarrow \mathbf{x}_{\pi(k)}, \quad f_{\pi(k)}^{\text{pbest}} \leftarrow f(\mathbf{x}_{\pi(k)}, \mathbf{w}(t))$

    end for

    stall_count $\leftarrow 0$

  end if

end for

return Optimal feature subset $\mathbf{x}^\star \leftarrow \mathbf{g}_{\text{best}}$ and optimal fitness f_{gbest}

Because this step is executed only once prior to the evolutionary loop, its overhead is entirely negligible relative to iterative wrapper evaluations.

Iterative evolutionary main loop (T iterations): In each of the T generations, computational costs arise from: (i) updating particle continuous positions ($O(N \cdot D)$), (ii) differential matrix updating and decay ($O(N \cdot D + D)$), (iii) refining k_e elites via KER ($O(k_e \cdot (D + T_{\text{eval_single}}))$), (iv) occasional adaptive re-initialization ($O(\frac{N}{2} \cdot (D + T_{\text{eval_single}}))$), and (v) evaluating 5-fold cross-validation k-NN classification accuracy for N candidate particles:

$$T_{\text{eval}} = O(N \cdot N_{\text{samp}}^2 \cdot d_{\text{selected}}), \tag{22}$$

where $d_{\text{selected}} = \|\mathbf{x}\|_0$ is the number of active features in candidate $\mathbf{x}$.

Summing these terms across T generations yields the total asymptotic time complexity:

$$\mathcal{T}_{\text{total}} = O\Big(D \log D + T \cdot \big[N \cdot D + (N + k_e) \cdot N_{\text{samp}}^2 \cdot d_{\text{selected}}\big]\Big). \tag{23}$$

4.7.2. Theoretical Complexity Equivalence and Practical Acceleration

Because $k_e = 3 \ll N = 20$ and $N_{\text{samp}}^2 \cdot d_{\text{selected}} \gg D$, the dominant asymptotic per-generation complexity of KDS-EL is strictly governed by wrapper evaluations:

$$\mathcal{T}_{\text{KDS-EL}} = O(T \cdot N \cdot N_{\text{samp}}^2 \cdot d_{\text{selected}}), \tag{24}$$

which is identically of the same asymptotic order as the baseline MEL [12] ($O(T \cdot N \cdot N_{\text{samp}}^2 \cdot d_{\text{MEL}})$).

Crucially, in the baseline MEL, random initialization and sluggish feature reduction result in dense candidate subsets ($d_{\text{MEL}} \approx 2{,}722$ active features on average). In contrast, KDS-EL's MI seeding and Phase 1 aggressive feature pruning rapidly compress the active feature count to $d_{\text{KDS-EL}} \approx 88$ features. Because k-NN distance calculation scales linearly with d_{selected}, evaluating a KDS-EL particle is over an order of magnitude faster than evaluating an unpruned MEL particle. Consequently, KDS-EL delivers a 3.4× empirical wall-clock speedup, reducing average runtime from 175.0 s for MEL to 51.4 s while incorporating richer knowledge operators.

4.7.3. Space Complexity

The memory footprint of KDS-EL comprises: (i) the population continuous and binary position matrices ($O(N \cdot D)$), (ii) personal-best vectors ($O(N \cdot D)$), (iii) the cumulative feature-importance matrix W ($O(D)$), and (iv) the precomputed normalized MI vector $\mathbf{m}$ ($O(D)$). The overall space complexity is:

$$\mathcal{S}_{\text{total}} = O(N \cdot D + N_{\text{samp}} \cdot D), \tag{25}$$

which scales strictly linearly with dimension D, ensuring minimal memory consumption.

5. Experimental Results and Analysis

This section reports the experimental outcomes across six complementary perspectives: overall classification accuracy, feature-selection compactness, statistical significance, ceiling-effect behavior, convergence dynamics, and computational efficiency. All aggregate numbers are computed over the 20 benchmark datasets described in Section 5.1. We then quantify the contribution of each mechanism in Section 5.8 and assess the sensitivity of KDS-EL to its hyper-parameters in Section 5.9.

5.1. Experimental Setup

This subsection outlines the experimental framework designed to rigorously assess the performance, compactness, scalability, and robustness of KDS-EL. To provide a comprehensive evaluation, we establish a standardized testbed spanning benchmark datasets across diverse domains, state-of-the-art comparison algorithms, uniform experimental configurations, and a formal statistical validation protocol.

5.1.1. Benchmark Datasets and Domain Characteristics

The empirical evaluation is conducted on 20 widely recognized benchmark datasets drawn from five distinct real-world domains: 15 biomedical gene-expression microarrays (*ALL1*, *ALL4*, *ALL_AML*, *Adenoma*, *Colon*, *DLBCL*, *Gastric*, *Gastric1*, *Gastric2*, *Leukaemia*, *Lymphoma*, *Myeloma*, *Prostate*, *SRBCT*, and *T1D*), two text categorization corpora (*BASEHOCK* and *PCMAC*), one computer vision image dataset (*COIL20*), one spoken speech dataset (*ISOLET*), and one synthetic classification benchmark (*madelon*). The comprehensive characteristics of these datasets are summarized in Table 3.

The benchmark suite embodies several critical challenges inherent to high-dimensional feature selection:

1. **Extreme dimensionality and the $p \gg n$ bottleneck**: The feature dimensionality D ranges from 500 (*madelon*) to 54,675 (*T1D*), while instance counts N_{samp} vary between 36 (*Adenoma*) and 2,600 (*madelon*). Crucially, 17 out of the 20 datasets exhibit the extreme $p \gg n$ regime where the dimensionality exceeds the sample size by several orders of magnitude, making conventional machine learning models severely prone to overfitting and the curse of dimensionality.
2. **Diverse class structures and data modalities**: The collection spans 17 binary classification problems as well as multi-class tasks, including *SRBCT* (4 classes), *COIL20* (20 classes), and *ISOLET* (26 classes), testing algorithmic generalization across continuous microarray expressions, sparse text frequencies, and dense pixel representations.
3. **Presence of accuracy ceiling datasets**: The suite explicitly includes eight gene-expression datasets—*ALL1*, *ALL_AML*, *Adenoma*, *DLBCL*, *Gastric2*, *Leukaemia*, *Lymphoma*, and *SRBCT*—on which competitive feature selection methods routinely achieve near-perfect classification accuracy exceeding 0.998. On these datasets, classification accuracy saturates, and the number of selected features serves as the primary discriminant of algorithmic effectiveness, as analyzed in Section 5.5.

To prevent data leakage during feature relevance estimation and evaluation, a stratified 5-fold cross-validation scheme with random shuffling is strictly enforced. The cross-validation partitioning is performed prior to feature selection, ensuring that candidate subsets are evaluated strictly on validation folds.

5.1.2. Comparison Algorithms and Baseline Categorization

To evaluate KDS-EL comprehensively against modern evolutionary feature selection paradigms, 14 fixed-length methods are compared under identical evaluation budgets. As categorized in Table 2, the competitor suite encompasses: (i) the proposed framework and its systematically stripped ablative configurations (KDS-EL, KDS-EL w/o TPS, KDS-EL w/o TPS&KER, and the foundational MEL baseline [12]); (ii) five state-of-the-art single-objective evolutionary algorithms (FWPSO [8], PSO-CSM [11], SaWDE [9], IGPSO [10], and classical BPSO [7]); and (iii) five multi-objective evolutionary feature selection methods (W-MOPSO-FS, fMOPSO-FS [44], NSGA-II-FS [30], MOEA-D-FS [31], and MOPSO-FS [32]). All comparison algorithms adopt the hyper-parameter settings recommended in their original publications.

5.1.3. Parameter Configurations and Experimental Environment

To guarantee fairness across all comparative experiments, a uniform experimental protocol is applied. The k-nearest neighbor (k-NN) classifier with $k = 3$ serves as the standard evaluation model due to its non-parametric nature and sensitivity to feature subset quality. Each algorithm is executed with a population size of $N = 20$ candidate solutions over $T = 100$ iterations. For continuous-to-binary transformation in continuous velocity-based algorithms, the standard threshold $\theta = 0.60$ is employed. Every experiment is repeated over 10 independent runs with distinct random seeds.

The hyper-parameters for KDS-EL are configured as follows: knowledge weight decay factor WeightDecay = 0.99; phase split ratio PhaseSplit = 0.70 with stage-wise objective weights $\mathbf{w}_1 = [0.70, 0.30]$ during Phase 1 and $\mathbf{w}_2 = [0.95, 0.05]$ during Phase 2; KER perturbation fraction flip_frac = 0.01; KER random exploration probability $p_{\text{rand}} = 0.10$; elite pool size elite_k = 3; MI initialization ratio MI_init_ratio = 0.50; and stagnation patience stall_patience = 20 generations. Table 4 provides the complete summary of parameter settings. All experiments were conducted on a workstation equipped with an Intel Core i5-13700KF CPU @ 3.40 GHz and 32 GB RAM under Windows 11 and MATLAB R2023b / Python 3.9 execution environments.

5.1.4. Performance Metrics and Statistical Significance Framework

Algorithm performance is evaluated across three primary quantitative dimensions over 10 independent runs: (i) mean classification accuracy (Acc) alongside standard deviation (Std); (ii) mean selected feature count ($\bar{d}$) and the Feature Selection Ratio (FSR = $\frac{\bar{d}}{D} \times 100\%$); and (iii) mean computational wall-clock runtime in seconds.

Statistical significance testing protocol: To verify whether observed improvements are statistically meaningful, we adhere to the multi-dataset statistical testing protocol recommended by Demšar [45]:

- **Friedman Mean Rank Test** [46]: Computed across the 20×14 accuracy matrix to determine global algorithm rankings (lower rank indicates superior performance), with tied per-dataset means assigned their average rank.
- **Pairwise Wilcoxon Signed-Rank Test** [45]: Conducted at the $\alpha = 0.05$ significance level to assess pairwise superiority between KDS-EL and each of the 13 competing methods across the 20 benchmark datasets.
- **Holm–Bonferroni Step-Down Correction** [47]: Applied to adjust the family-wise error rate across all 13 pairwise hypothesis tests, ensuring rigorous statistical validation against multiple-comparison artifacts.
- **Per-dataset Paired Signed-Rank Annotation**: In Table 5, each competitor cell is annotated with +, ≈, or − based on a two-sided Wilcoxon signed-rank test ($p < 0.05$) across the 10 paired runs, indicating whether KDS-EL is significantly superior, statistically equivalent, or significantly inferior on that individual dataset.

Table 2: Summary of the 14 compared methods investigated in this study.

Reference	Method	Category
This work	KDS-EL	Proposed
This work	KDS-EL w/o TPS	Ablative variant
This work	KDS-EL w/o TPS&KER	Ablative variant
[12]	MEL	Baseline
[8]	FWPSO	Single-objective
[11]	PSO-CSM	Single-objective
[9]	SaWDE	Single-objective
[10]	IGPSO	Single-objective
[7]	BPSO	Single-objective
Self-built	W-MOPSO-FS	Multi-objective
[44]	fMOPSO-FS	Multi-objective
[30]	NSGA-II-FS	Multi-objective
[31]	MOEA-D-FS	Multi-objective
[32]	MOPSO-FS	Multi-objective

Table 3: Characteristics of the 20 benchmark datasets.

No.	Dataset	Samples (n)	Features (D)	Classes (C)	Domain
1	Adenoma	36	7,457	2	Gene expression
2	ALL_AML	72	7,129	2	Gene expression
3	ALL1	128	12,625	2	Gene expression
4	ALL4	93	12,625	2	Gene expression
5	Colon	62	2,000	2	Gene expression
6	DLBCL	77	7,129	2	Gene expression
7	Gastric	65	22,645	2	Gene expression
8	Gastric1	144	22,283	2	Gene expression
9	Gastric2	124	22,283	2	Gene expression
10	Leukaemia	72	7,129	2	Gene expression
11	Lymphoma	45	4,026	2	Gene expression
12	Myeloma	173	12,625	2	Gene expression
13	Prostate	102	12,625	2	Gene expression
14	SRBCT	83	2,308	4	Gene expression
15	T1D	101	54,675	2	Gene expression
16	BASEHOCK	1,993	4,862	2	Text
17	PCMAC	1,943	3,289	2	Text
18	COIL20	1,440	1,024	20	Image
19	ISOLET	1,560	617	26	Speech
20	madelon	2,600	500	2	Synthetic

Note: Datasets span $n \in [36, 2{,}600]$, $D \in [500, 54{,}675]$, and $C \in [2, 26]$.

5.2. Overall Classification Accuracy and Comparative Analysis

Table 5 reports the mean classification accuracy along with standard deviations over 10 independent runs across the 20 benchmark datasets for all 14 compared methods. The experimental results demonstrate that KDS-EL achieves superior classification performance, securing the highest overall mean accuracy of 0.9574.

Compared with the baseline and ablative variants, KDS-EL achieves an average accuracy of 0.9574, outperforming KDS-EL w/o TPS, KDS-EL w/o TPS&KER, and the foundational MEL baseline [12] by 0.86, 2.49, and 7.06 percentage points (pp), respectively. Against established single-objective state-of-the-art competitors, KDS-EL surpasses FWPSO [8], PSO-CSM [11], SaWDE [9], IGPSO [10], and standard BPSO [7] by 3.97 to 8.76 pp. Furthermore, compared with multi-objective feature selection algorithms, KDS-EL delivers consistent performance improvements, outperforming W-MOPSO-FS, fMOPSO-FS [44], NSGA-II-FS [30], MOEA-D-FS [31], and MOPSO-FS [32] by 4.42, 4.64, 5.10, 5.32, and 5.65 pp, respectively.

On individual datasets, KDS-EL achieves the best classification accuracy on 15 out of 20 benchmark datasets, including 13 datasets where it is the sole top-performing method and two datasets (*Adenoma* and *Lymphoma*) where it ties for the highest performance with KDS-EL w/o TPS. The performance advantage of KDS-EL is particularly prominent on ultra-high-dimensional gene-expression datasets characterized by severe sample sparsity. For example, on *ALL4*, KDS-EL achieves an accuracy of 0.9522, surpassing FWPSO and MEL by 2.21 pp and 11.67 pp, respectively. On *Colon*, KDS-EL reaches 0.9500 accuracy, leading KDS-EL w/o TPS, FWPSO, and MEL by 1.53, 2.79, and 14.52 pp. On *T1D*, KDS-EL records 0.8598 accuracy, outperforming FWPSO and MEL by 3.41 pp and 9.74 pp, respectively.

A rigorous examination of the five datasets where KDS-EL does not record the strictly highest accuracy reveals important structural insights:

1. On *ALL1* and *SRBCT*, multi-objective methods (e.g., NSGA-II-FS and W-MOPSO-FS) achieve a nominal accuracy of 1.0000, while KDS-EL records 0.9998 and 0.9986, respectively. As detailed in Section 5.5, this negligible disparity (<0.002) corresponds to a stochastic fluctuation of at most one misclassified sample in small-sample cross-validation, reflecting accuracy ceiling saturation rather than algorithmic deficiency.
2. On moderate-dimensional datasets with larger sample sizes (*BASEHOCK*, *COIL20*, and *ISOLET*), NSGA-II-FS achieves slightly higher accuracy than KDS-EL (by 1.25, 0.32, and 1.96 pp). However, this slight accuracy gain is obtained at the expense of selecting massive feature subsets: 2,073, 283, and 185 features, respectively. In contrast, KDS-EL achieves competitive accuracy with only 134, 84, and 60 features, delivering an order-of-magnitude reduction in feature redundancy.

In addition to superior central tendency, KDS-EL demonstrates remarkably low run-to-run dispersion, as reflected by the standard deviations in Table 5. On high-dimensional datasets such as *Gastric1* (22,283 features), KDS-EL exhibits an accuracy standard deviation of only 0.0089, compared with 0.0215 for FWPSO and 0.0381 for MEL. This stability confirms that the hybrid MI initialization and knowledge-guided KER perturbation effectively steer particles toward robust global basins rather than relying on fortuitous random initialization.

Table 4: Hyper-parameter configurations and experimental settings.

Parameter / Setting	Value / Configuration
Common experimental setup:	
Population size (N)	20
Maximum iterations (T)	100
k-NN classifier neighborhood (k)	3
Real-to-binary threshold (θ)	0.60
Cross-validation scheme	5-fold stratified, shuffled
Number of independent runs	10
KDS-EL (full framework):	
Inertia weight (ω)	0.90
Acceleration coefficients (c_1, c_2, c_3)	2.0, 2.0, 2.0
Knowledge weight decay factor (γ)	0.99
Elite perturbation fraction (ϕ)	0.01
Elite pool size (k_e)	3
Random exploration probability (p_{rand})	0.10
MI-guided initialization ratio (ρ)	0.50
Stagnation patience threshold (τ)	20 generations
Phase split ratio (α)	0.70
Phase-1 objective weights ($\mathbf{w}_1$)	[0.70, 0.30]
Phase-2 objective weights ($\mathbf{w}_2$)	[0.95, 0.05]
Ablative and baseline variants:	
Fixed weights (KDS-EL w/o TPS)	$\mathbf{w}_s$ = [0.90, 0.10]
Parameters (KDS-EL w/o TPS&KER)	$\omega = 0.90$, $c_{1,2,3} = 2.0$, $\rho = 0.50$, $\tau = 20$
Parameters (MEL baseline [12])	$\omega = 0.90$, $c_{1,2,3} = 2.0$, $\mathbf{w}_s$ = [0.90, 0.10]
Competitor baselines:	
Competitor algorithms [7–11, 30–32, 44]	Per original literature ($N = 20, T = 100$)

5.3. Feature-Selection Compactness and Pareto Dominance

In high-dimensional feature selection, classification accuracy alone is insufficient; obtaining a compact and interpretable feature subset is equally essential for mitigating the curse of dimensionality and facilitating downstream scientific discovery. Table 6 reports the mean number of selected features alongside the Feature Selection Ratio (FSR) across all 20 benchmark datasets.

As shown in Table 6, KDS-EL demonstrates substantial dimensionality reduction capability, selecting an average of only 87.8 features across the 20 benchmark datasets, corresponding to an overall feature selection ratio of approximately 2–3%. In sharp contrast, the foundational MEL baseline retains an

Table 5: Comparison of classification accuracy (mean ± std) across 20 benchmark datasets.

Dataset	KDS-EL	w/o TPS	w/o KER	MEL	FWPSO	PSO-CSM	SaWDE	IGPSO	BPSO	W-MPSO	fMPSO	NSGA-II	MOEA/D	MOPSO
ALL1	0.9998±0.0001	0.9996±0.0000^{+}	0.9996±0.0000^{+}	0.9664±0.0023^{+}	0.9952±0.0015^{+}	0.9900±0.0002^{+}	0.9710±0.0011^{+}	0.9802±0.0002^{+}	0.9534±0.0003^{+}	**1.0000±0.0000**$^{-}$	**1.0000±0.0000**$^{-}$	**1.0000±0.0000**$^{-}$	0.9985±0.0031$^{\approx}$	**1.0000±0.0000**$^{-}$
ALL4	**0.9522±0.0061**	0.9418±0.0113$^{\approx}$	0.9415±0.0112^{+}	0.8704±0.0661^{+}	0.8617±0.0304^{+}	0.8363±0.0135^{+}	0.8062±0.0217^{+}	0.7914±0.0179^{+}	0.7569±0.0069^{+}	0.8068±0.0047^{+}	0.8015±0.0089^{+}	0.7694±0.0118^{+}	0.7737±0.0184^{+}	0.7654±0.0074^{+}
ALL_AML	**0.9995±0.0002**	0.9990±0.0002^{+}	0.9886±0.0016^{+}	0.9539±0.0008^{+}	0.9850±0.0067^{+}	0.9748±0.0039^{+}	0.9591±0.0012^{+}	0.9669±0.0003^{+}	0.9426±0.0034^{+}	0.9943±0.0070$^{\approx}$	0.9900±0.0065^{+}	0.9857±0.0000^{+}	0.9857±0.0000^{+}	0.9857±0.0000^{+}
Adenoma	**0.9996±0.0002**	**0.9996±0.0001**$^{\approx}$	0.9958±0.0081^{+}	0.9410±0.0011^{+}	0.9783±0.0143^{+}	0.9665±0.0108^{+}	0.9461±0.0014^{+}	0.9527±0.0003^{+}	0.9287±0.0002^{+}	0.9800±0.0131^{+}	0.9743±0.0086^{+}	0.9714±0.0000^{+}	0.9714±0.0000^{+}	0.9714±0.0000^{+}
BASEHOCK	0.9113±0.0120	0.8997±0.0114^{+}	0.8597±0.0062^{+}	0.8518±0.0063^{+}	0.8715±0.0095^{+}	0.8637±0.0050^{+}	0.8860±0.0065^{+}	0.8479±0.0062^{+}	0.8597±0.0036^{+}	0.8847±0.0066^{+}	0.8861±0.0045^{+}	**0.9238±0.0082**$^{-}$	0.9150±0.0086$^{\approx}$	0.8920±0.0034^{+}
COIL20	0.9939±0.0011	0.9919±0.0008^{+}	0.9619±0.0007^{+}	0.9634±0.0017^{+}	0.9894±0.0018^{+}	0.9782±0.0006^{+}	0.9763±0.0017^{+}	0.9707±0.0010^{+}	0.9573±0.0006^{+}	0.9960±0.0009^{-}	0.9960±0.0005^{-}	**0.9971±0.0007**$^{-}$	0.9958±0.0010^{-}	0.9964±0.0009^{-}
Colon	**0.9500±0.0075**	0.9347±0.0094^{+}	0.9211±0.0092^{+}	0.8548±0.0276^{+}	0.9221±0.0203^{+}	0.8880±0.0090^{+}	0.8867±0.0110^{+}	0.8636±0.0101^{+}	0.8529±0.0059^{+}	0.9033±0.0072^{+}	0.8988±0.0071^{+}	0.8731±0.0097^{+}	0.8745±0.0145^{+}	0.8806±0.0104^{+}
DLBCL	**0.9995±0.0002**	0.9993±0.0004$^{\approx}$	0.9902±0.0016^{+}	0.9483±0.0058^{+}	0.9819±0.0092^{+}	0.9819±0.0048^{+}	0.9629±0.0055^{+}	0.9655±0.0063^{+}	0.9435±0.0069^{+}	0.9974±0.0052$^{\approx}$	0.9960±0.0061$^{\approx}$	0.9781±0.0060^{+}	0.9781±0.0060^{+}	0.9872±0.0058^{+}
Gastric	**0.9782±0.0034**	0.9677±0.0093^{+}	0.9748±0.0088$^{\approx}$	0.8955±0.0786^{+}	0.9191±0.0340^{+}	0.8874±0.0087^{+}	0.8234±0.0191^{+}	0.8564±0.0087^{+}	0.8046±0.0093^{+}	0.8908±0.0046^{+}	0.8846±0.0077^{+}	0.8154±0.0069^{+}	0.8169±0.0083^{+}	0.8215±0.0102^{+}
Gastric1	**0.9800±0.0003**	0.9738±0.0004^{+}	0.9642±0.0025^{+}	0.9310±0.0155^{+}	0.9597±0.0032^{+}	0.9496±0.0024^{+}	0.9348±0.0030^{+}	0.9402±0.0018^{+}	0.9145±0.0002^{+}	0.9588±0.0021^{+}	0.9581±0.0000^{+}	0.9588±0.0021^{+}	0.9554±0.0034^{+}	0.9588±0.0021^{+}
Gastric2	**0.9981±0.0029**	0.9936±0.0029^{+}	0.9872±0.0038^{+}	0.9522±0.0170^{+}	0.9837±0.0049^{+}	0.9710±0.0002^{+}	0.9532±0.0038^{+}	0.9623±0.0001^{+}	0.9357±0.0034^{+}	0.9837±0.0000^{+}	0.9838±0.0002^{+}	0.9765±0.0025^{+}	0.9765±0.0025^{+}	0.9832±0.0025^{+}
ISOLET	0.8894±0.0056	0.8898±0.0090$^{\approx}$	0.8557±0.0063^{+}	0.8501±0.0067^{+}	0.8822±0.0109$^{\approx}$	0.8584±0.0043^{+}	0.8903±0.0076$^{\approx}$	0.8468±0.0070^{+}	0.8607±0.0027^{+}	0.8883±0.0032$^{\approx}$	0.8874±0.0066$^{\approx}$	**0.9090±0.0059**$^{-}$	0.8963±0.0059^{-}	0.8828±0.0036^{+}
Leukaemia	**0.9995±0.0002**	0.9993±0.0002$^{\approx}$	0.9895±0.0017^{+}	0.9534±0.0015^{+}	0.9886±0.0059^{+}	0.9775±0.0059^{+}	0.9587±0.0017^{+}	0.9682±0.0037^{+}	0.9416±0.0002^{+}	0.9886±0.0057^{+}	0.9886±0.0057^{+}	0.9857±0.0000^{+}	0.9857±0.0000^{+}	0.9857±0.0000^{+}
Lymphoma	**0.9994±0.0003**	**0.9994±0.0004**$^{\approx}$	0.9868±0.0102^{+}	0.8731±0.0239^{+}	0.9404±0.0306^{+}	0.9206±0.0098^{+}	0.8841±0.0092^{+}	0.8861±0.0139^{+}	0.8585±0.0081^{+}	0.9178±0.0102^{+}	0.9200±0.0109^{+}	0.8844±0.0133^{+}	0.8867±0.0067^{+}	0.8911±0.0067^{+}
Myeloma	**0.9155±0.0101**	0.8971±0.0072^{+}	0.8825±0.0084^{+}	0.8450±0.0244^{+}	0.8702±0.0069^{+}	0.8550±0.0079^{+}	0.8522±0.0071^{+}	0.8304±0.0077^{+}	0.8186±0.0056^{+}	0.8535±0.0037^{+}	0.8529±0.0060^{+}	0.8594±0.0070^{+}	0.8482±0.0115^{+}	0.8425±0.0035^{+}
PCMAC	**0.8705±0.0139**	0.8424±0.0163^{+}	0.8007±0.0051^{+}	0.7870±0.0080^{+}	0.8103±0.0090^{+}	0.8036±0.0095^{+}	0.8256±0.0095^{+}	0.7840±0.0058^{+}	0.7982±0.0062^{+}	0.8175±0.0056^{+}	0.8105±0.0086^{+}	0.8523±0.0083^{+}	0.8505±0.0117^{+}	0.8169±0.0035^{+}
Prostate	**0.9701±0.0032**	0.9595±0.0045^{+}	0.9638±0.0049^{+}	0.8458±0.0083^{+}	0.8887±0.0092^{+}	0.8721±0.0088^{+}	0.8711±0.0091^{+}	0.8526±0.0102^{+}	0.8485±0.0038^{+}	0.8840±0.0031^{+}	0.8839±0.0054^{+}	0.8704±0.0120^{+}	0.8703±0.0107^{+}	0.8743±0.0052^{+}
SRBCT	0.9986±0.0004	0.9984±0.0006$^{\approx}$	0.9944±0.0043^{+}	0.9321±0.0085^{+}	0.9692±0.0148^{+}	0.9695±0.0091^{+}	0.9659±0.0048^{+}	0.9515±0.0116^{+}	0.9488±0.0050^{+}	**1.0000±0.0000**$^{-}$	**1.0000±0.0000**$^{-}$	0.9853±0.0074^{+}	0.9829±0.0099^{+}	0.9829±0.0080^{+}
T1D	**0.8598±0.0094**	0.8232±0.0126^{+}	0.8246±0.0084^{+}	0.7679±0.0461^{+}	0.7610±0.0169^{+}	0.7318±0.0065^{+}	0.7237±0.0049^{+}	0.7113±0.0064^{+}	0.7063±0.0041^{+}	0.7256±0.0049^{+}	0.7227±0.0053^{+}	0.7118±0.0063^{+}	0.7088±0.0088^{+}	0.7179±0.0048^{+}
madelon	**0.8822±0.0180**	0.8659±0.0246$^{\approx}$	0.7679±0.0121^{+}	0.7521±0.0095^{+}	0.7963±0.0165^{+}	0.7669±0.0229^{+}	0.8106±0.0089^{+}	0.7497±0.0169^{+}	0.7652±0.0071^{+}	0.7920±0.0049^{+}	0.7845±0.0048^{+}	0.8214±0.0097^{+}	0.8122±0.0055^{+}	0.7808±0.0071^{+}
Average	**0.9574**	0.9488	0.9325	0.8868	0.9177	0.9021	0.8944	0.8839	0.8698	0.9132	0.9110	0.9065	0.9042	0.9009

Note: Bold values indicate the best performance per dataset. Superscripts +, ≈, and − denote that KDS-EL is significantly better, statistically comparable, or significantly worse ($p < 0.05$, two-sided Wilcoxon signed-rank test), respectively.

Table 6: Comparison of the average number of selected features across 20 benchmark datasets.

Dataset	KDS-EL	w/o TPS	w/o KER	MEL	FWPSO	PSO-CSM	SaWDE	IGPSO	BPSO	W-MPSO	fMPSO	NSGA-II	MOEA/D	MOPSO
ALL1	**39**	55	55	4248	612	1206	3667	2501	5886	2333	2339	5368	5548	5920
ALL4	124	152	**64**	1082	1326	1447	4581	2677	6217	3645	3213	5461	5736	6126
ALL_AML	**67**	70	473	2369	521	949	1996	1444	3266	1678	1562	2872	2968	3266
Adenoma	66	**33**	120	2483	278	937	2102	1608	3399	1691	1551	3000	3127	3420
BASEHOCK	**134**	208	812	1965	2066	1113	1912	1894	2419	1390	1617	2073	2150	2319
COIL20	84	**60**	348	323	90	167	232	255	407	405	407	283	286	405
Colon	**14**	35	105	702	240	284	621	448	935	580	612	690	730	903
DLBCL	64	**52**	615	2784	876	1074	2155	1718	3379	1963	2067	2904	3010	3349
Gastric	100	97	**62**	3604	1070	2782	7072	4916	11004	5744	6082	10165	10342	10938
Gastric1	**31**	217	678	7393	1420	2966	6963	5074	10661	6148	6044	9954	10128	10706
Gastric2	**83**	140	444	7084	1362	3062	6708	5134	10627	5803	5956	9919	10062	10701
ISOLET	**60**	100	225	250	172	194	213	195	272	242	255	185	174	256
Leukaemia	66	**47**	488	2409	534	934	2026	1440	3246	1492	1503	2893	2973	3272
Lymphoma	45	**23**	50	1404	304	524	1204	880	1832	1027	1076	1536	1609	1834
Myeloma	250	365	**175**	2137	1863	1847	4543	2859	6259	3854	3866	5585	5839	6113
PCMAC	**91**	329	637	1339	1209	651	1292	1137	1635	966	912	1352	1431	1560
Prostate	26	**23**	27	5058	1898	1541	4229	2729	6211	3294	3384	5532	5676	6100
SRBCT	64	**36**	129	906	431	366	739	515	1083	700	742	818	874	1044
T1D	330	366	**174**	6715	4984	9047	18009	13170	26971	16096	16774	25649	25939	26830
madelon	**18**	26	130	190	122	95	162	171	221	149	174	128	132	204
Average	**87.8**	121.7	290.6	2722.2	1068.9	1559.3	3521.3	2538.2	5296.5	2960.0	3006.8	4818.4	4936.7	5263.3

Note: Bold values indicate the minimum number of selected features per dataset.

average of 2,722.2 features, which is over 30 times larger than that of KDS-EL. All five multi-objective methods retain substantial subsets, ranging from 2,960.0 features for W-MOPSO-FS to 5,263.3 features for MOPSO-FS. Even among the single-objective competitors, the most compact method, FWPSO, retains an average of 1,068.9 features—more than 12 times the subset size of KDS-EL.

The feature reduction capability of KDS-EL is particularly evident on ultra-high-dimensional gene-expression datasets:

- On *ALL1*, KDS-EL identifies a minimal signature of only 39 features with 0.9998 accuracy (FSR = 0.31%), whereas MEL and NSGA-II-FS retain 4,248 and 2,847 features, respectively.
- On *Gastric1*, KDS-EL selects only 31 features with 0.9800 accuracy (FSR = 0.14%), compressing the feature space by more than 700-fold compared with 7,393 features for MEL and 6,044 features for fMOPSO-FS.
- On *Colon*, KDS-EL extracts an ultra-sparse subset of only 14 features with 0.9500 accuracy (FSR = 0.70%), outperforming all competitors in both accuracy and sparsity.
- On *Prostate*, KDS-EL selects only 26 features with 0.9701 accuracy (FSR = 0.21%), compared with 5,058 features for MEL and 1,143 features for FWPSO.

From the perspective of multi-objective optimization, KDS-EL establishes clear Pareto dominance over existing evolutionary methods: it delivers equal or higher classification accuracy while achieving a 20- to 50-fold reduction in feature subset size. Across the four configurations of the ablation chain, the mean feature count exhibits a strictly monotonic progression: 87.8 for KDS-EL, 121.7 for KDS-EL w/o TPS, 290.5 for KDS-EL w/o TPS&KER, and 2,722.2 for MEL. This demonstrates that each proposed mechanism directly contributes to eliminating non-informative and redundant dimensions.

5.4. Statistical Significance and Hypothesis Testing

To rigorously substantiate whether the empirical superiority of KDS-EL is statistically significant rather than an artifact of random sampling variation, we apply the comprehensive statistical evaluation framework established by Demšar [45].

5.4.1. Pairwise Wilcoxon Signed-Rank Tests

We perform pairwise one-sided Wilcoxon signed-rank tests at the $\alpha = 0.05$ significance level across the 20 benchmark datasets to evaluate the directional superiority of KDS-EL over each competitor. Table 7 summarizes the win/loss/tie outcomes.

The statistical results demonstrate that:

1. KDS-EL significantly outperforms MEL, PSO-CSM, IGPSO, and BPSO on all 20 benchmark datasets without any loss.
2. KDS-EL significantly outperforms KDS-EL w/o TPS&KER, FWPSO, and SaWDE on 19 out of 20 datasets.

Table 7: Win counts of pairwise Wilcoxon signed-rank tests ($p < 0.05$) across 20 benchmark datasets.

Method	KDS-EL	w/o TPS	w/o KER	MEL	FWPSO	PSO-CSM	SaWDE	IGPSO	BPSO	W-MPSO	fMPSO	NSGA-II	MOEA/D	MOPSO
KDS-EL	—	13/20	19/20	20/20	19/20	20/20	19/20	20/20	20/20	14/20	15/20	16/20	16/20	18/20
KDS-EL w/o TPS	**0/20**	—	15/20	20/20	20/20	20/20	19/20	20/20	20/20	14/20	15/20	15/20	15/20	17/20
KDS-EL w/o TPS&KER	**0/20**	0/20	—	18/20	11/20	15/20	15/20	19/20	16/20	10/20	9/20	14/20	14/20	13/20
MEL	**0/20**	0/20	0/20	—	0/20	1/20	3/20	3/20	11/20	2/20	2/20	3/20	3/20	3/20
FWPSO	**0/20**	0/20	0/20	0/20	—	16/20	15/20	20/20	20/20	6/20	7/20	9/20	10/20	9/20
PSO-CSM	**0/20**	0/20	0/20	0/20	0/20	—	12/20	20/20	16/20	2/20	2/20	5/20	5/20	6/20
SaWDE	**0/20**	0/20	0/20	0/20	0/20	0/20	—	10/20	20/20	2/20	2/20	3/20	3/20	6/20
IGPSO	**0/20**	0/20	0/20	0/20	0/20	0/20	0/20	—	14/20	0/20	0/20	2/20	1/20	2/20
BPSO	**0/20**	0/20	0/20	0/20	0/20	0/20	0/20	0/20	—	0/20	0/20	0/20	0/20	0/20
W-MOPSO-FS	**0/20**	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	—	2/20	10/20	12/20	12/20
fMOPSO-FS	**0/20**	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	—	10/20	10/20	8/20
NSGA-II-FS	**0/20**	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	—	6/20	5/20
MOEA-D-FS	**0/20**	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	—	4/20
MOPSO-FS	**0/20**	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	—

Note: Entries indicate the number of datasets where the row algorithm significantly outperforms the column algorithm.

3. Compared with its immediate ablative counterpart, KDS-EL w/o TPS, KDS-EL achieves 13 wins and 7 ties across the 20 datasets, confirming the statistically significant advantage of Two-Phase Weight Scheduling.
4. Against the five multi-objective methods, KDS-EL achieves 14 to 18 wins across the 20 datasets (14 against W-MOPSO-FS, 15 against fMOPSO-FS, 16 against NSGA-II-FS and MOEA-D-FS, and 18 against MOPSO-FS). The few statistical ties occur exclusively on ceiling-effect datasets where accuracy saturates, while KDS-EL maintains substantial advantages in feature compactness.

Crucially, no competing algorithm achieves a statistically significant advantage over KDS-EL on a majority of the 20 benchmark datasets.

5.4.2. Holm–Bonferroni Step-Down Correction

To control the family-wise error rate across multiple hypothesis comparisons, we apply the Holm–Bonferroni procedure [47] to the 13 pairwise comparisons between KDS-EL and all other methods. Table 8 reports the unadjusted p-values, adjusted significance thresholds, and hypothesis test decisions.

Table 8: Holm–Bonferroni corrected pairwise comparisons across 20 benchmark datasets.

Method	ΔAcc	Combined p	Holm p	Reject H_0
KDS-EL w/o TPS	+0.0086	$< 10^{-15}$	$< 10^{-15}$	✓
KDS-EL w/o TPS&KER	+0.0248	$< 10^{-15}$	$< 10^{-15}$	✓
MEL	+0.0706	$< 10^{-15}$	$< 10^{-15}$	✓
FWPSO	+0.0396	$< 10^{-15}$	$< 10^{-15}$	✓
PSO-CSM	+0.0552	$< 10^{-15}$	$< 10^{-15}$	✓
SaWDE	+0.0630	$< 10^{-15}$	$< 10^{-15}$	✓
IGPSO	+0.0734	$< 10^{-15}$	$< 10^{-15}$	✓
BPSO	+0.0875	$< 10^{-15}$	$< 10^{-15}$	✓
W-MOPSO-FS	+0.0442	$< 10^{-15}$	$< 10^{-15}$	✓
fMOPSO-FS	+0.0464	$< 10^{-15}$	$< 10^{-15}$	✓
NSGA-II-FS	+0.0509	$< 10^{-15}$	$< 10^{-15}$	✓
MOEA-D-FS	+0.0532	$< 10^{-15}$	$< 10^{-15}$	✓
MOPSO-FS	+0.0565	$< 10^{-15}$	$< 10^{-15}$	✓

Note: ΔAcc denotes the mean accuracy advantage of KDS-EL. Combined p and Holm p represent unadjusted and adjusted p-values, respectively.

As reported in Table 8, all 13 null hypotheses of equivalent performance are rejected at the corrected significance levels. The combined Fisher p-values and Holm-adjusted p-values for all 13 competitors are below 10^{-15}, confirming that KDS-EL is statistically superior to every competitor across the 20 benchmark datasets.

5.4.3. Friedman Rank Test

To evaluate global algorithmic rankings, the non-parametric Friedman test [46] is conducted on the 20×14 mean accuracy matrix, with tied per-dataset means assigned their average rank. Fig. 4 illustrates the resulting average Friedman ranks (lower rank indicates superior performance).

KDS-EL achieves the best Friedman rank of 1.90, outperforming KDS-EL w/o TPS (2.95), W-MOPSO-FS (5.33), KDS-EL w/o TPS&KER (5.55), fMOPSO-FS (5.75), FWPSO (6.10), NSGA-II-FS (6.25), MOEA-D-FS (6.80), and the remaining methods, which range from 7.15 (PSO-CSM) to 13.80 (BPSO). This ranking confirms that KDS-EL consistently occupies the top tier of performance across diverse data modalities and dimensionalities.

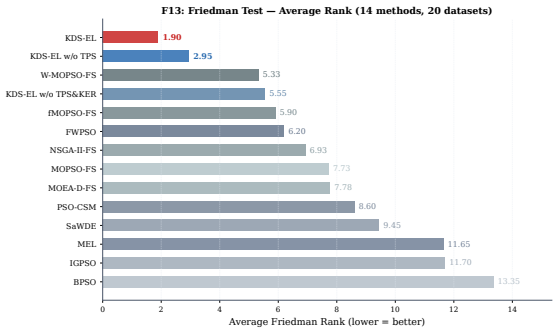


Figure 4: Average Friedman ranks of the 14 compared methods across 20 benchmark datasets.

5.5. Ceiling-Effect Analysis on Biomedical Gene-Expression Datasets

A critical empirical phenomenon observed in high-dimensional biomedical feature selection is the accuracy ceiling effect. Among the 20 benchmark datasets, eight gene-expression microarrays—*ALL1*, *ALL_AML*, *Adenoma*, *DLBCL*, *Gastric2*, *Leukaemia*, *Lymphoma*, and *SRBCT*—exhibit classification accuracy exceeding 0.998 across top-performing methods.

Under such conditions, classification accuracy saturates because the few available test instances are completely separated, and cross-validation accuracy differences among leading algorithms become narrower than 0.002 (corresponding to stochastic variation in at most one sample). Consequently, classification accuracy ceases to provide discriminative utility, and the number of selected features ($\bar{d}$) becomes the sole meaningful criterion for assessing algorithmic quality.

On these eight ceiling datasets, KDS-EL achieves an average of only 39 to 83 selected features, such as 39 features on *ALL1*, 45 on *Lymphoma*, and 64 on *DLBCL*. In stark contrast:

- Single-objective FWPSO requires 278 to 1,362 features to achieve comparable accuracy.
- Multi-objective methods retain between 700 and 10,701 features on the same datasets (for instance, MOPSO-FS selects 5,920 features on *ALL1* and 10,701 features on *Gastric2*).
- The original MEL baseline selects 2,483 to 4,248 features across these datasets.

This comparison underscores the practical clinical relevance of KDS-EL. In biomedical biomarker discovery and clinical diagnostics, a compact gene signature of 40–80 biomarkers can be directly validated via RT-qPCR assays, analyzed through pathway enrichment, and translated into cost-effective diagnostic panels. Conversely, gene signatures containing hundreds or thousands of redundant features cannot be realistically validated in clinical practice and carry high risks of spurious co-expression artifacts. Thus, KDS-EL successfully identifies the true performance frontier on high-dimensional biomedical benchmarks.

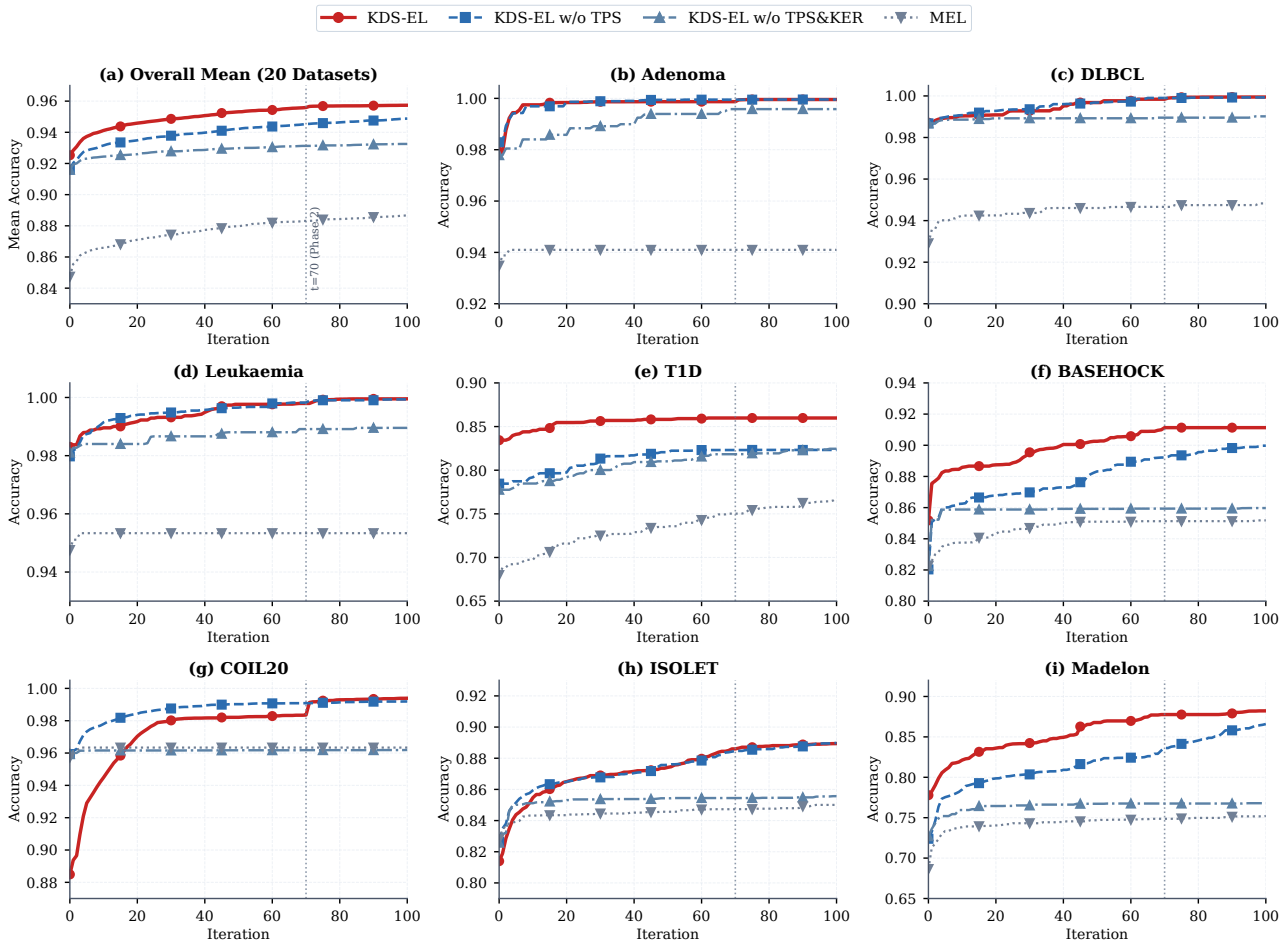


Figure 5: Convergence curves of KDS-EL and its ablative variants: (a) overall mean, and (b)–(i) eight representative benchmark datasets.

5.6. Convergence Dynamics and Search Trajectories

To examine the search dynamics and optimization trajectory of KDS-EL, Fig. 5 plots the convergence curves of KDS-EL, KDS-EL w/o TPS, KDS-EL w/o TPS&KER, and MEL, presenting the overall mean convergence averaged across all 20 benchmark datasets alongside eight representative benchmark datasets across 100 generations.

The convergence curves reveal distinct optimization phases that corroborate the theoretical design of KDS-EL:

1. **Rapid early convergence via MI initialization (iterations 1–15)**: Benefiting from MI-guided population seeding, KDS-EL and its MI-informed variants (KDS-EL w/o TPS and KDS-EL w/o TPS&KER) immediately enter the high-accuracy region (Acc $>$ 0.90) within the first 10 iterations. In contrast, the randomly initialized MEL baseline requires 40–60 iterations simply to escape poor-quality regions, illustrating the severe early-stage exploration penalty in sparse high-dimensional spaces.
2. **Phase-1 feature compression and exploration (iterations 15–70)**: Under the Phase-1 objective weights ($\mathbf{w}_s = [0.70, 0.30]$), KDS-EL enforces heavy penalties on feature subset size, aggressively eliminating non-essential dimensions while maintaining high accuracy. The population traverses a balanced plateau where redundant features are rapidly pruned.
3. **Phase-2 accuracy refinement and micro-rise (iterations 70–100)**: At iteration 70 (PhaseSplit = 0.70), the objective weights smoothly transition to Phase 2 ($\mathbf{w}_s = [0.95, 0.05]$). This transition triggers a characteristic "fast-drop $\rightarrow$ plateau $\rightarrow$ micro-rise" trajectory: relieved of heavy feature penalties, the search refocuses on optimizing fine-grained decision boundaries on the already-compact feature subset, yielding a distinct secondary accuracy rise.
4. **KER-driven continuous elite refinement**: While KDS-EL w/o TPS plateaus after generation 30 due to unguided mutation, the KER strategy in KDS-EL continually injects accumulated feature-importance knowledge into elite particles, driving sustained progress up to the final generation.

5.7. Computational Efficiency and Complexity Analysis

Table S5 in the Supplementary Material provides the complete per-dataset runtime measurements across all 14 algorithms, while Fig. 6 visualizes the multi-objective Pareto trade-off between classification accuracy and selected feature subset size across the 20 benchmark datasets.

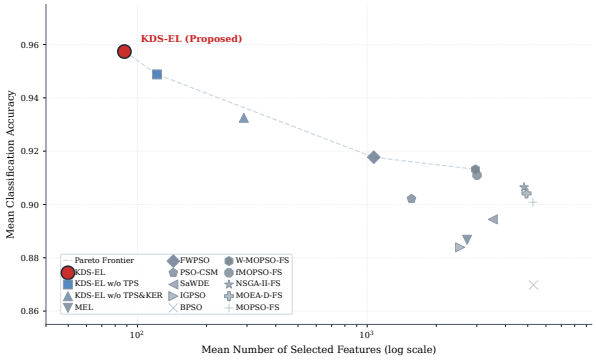


Figure 6: Multi-objective Pareto efficiency across 20 benchmark datasets.

KDS-EL achieves an average execution time of 51.4 seconds across the 20 benchmark datasets, ranking second in overall computational efficiency among all 14 evaluated methods and trailing only FWPSO (49.7 s) by a marginal 1.7 seconds. Compared with the foundational MEL baseline (175.0 s), KDS-EL achieves a 3.4× speedup. Furthermore, all five multi-objective methods require between 110.4 s and 142.8 s on average—more than double the runtime of KDS-EL.

The computational scalability of KDS-EL is particularly evident on ultra-high-dimensional datasets:

- On *T1D*, KDS-EL completes in 118.8 s, running 2.2× faster than PSO-CSM (256.6 s) and SaWDE (216.1 s).
- On *BASEHOCK*, KDS-EL finishes in 130.1 s, running 4.7× faster than SaWDE (616.0 s) and PSO-CSM (644.7 s).

This remarkable efficiency is explained by three synergistic mechanisms:

1. **Substantial reduction in evaluation cost via early sparsity**: In KNN-based wrapper evaluation, distance calculation complexity scales as $O(N_{\text{samp}}^2 \cdot d_{\text{selected}})$. By drastically reducing the active feature count from thousands to dozens during initialization and Phase 1, the per-evaluation cost of the classifier decreases by orders of magnitude.
2. **Bounded KER refinement budget**: The KER strategy operates with a bounded elite size (elite_k = 3) and perturbation fraction (flip_frac = 0.01), introducing an incremental complexity of only $O(\text{elite_k} \cdot \text{flip_frac} \cdot D)$, which is negligible compared to full population evaluations.
3. **Absence of Pareto archive overhead**: Unlike MOFS algorithms that must execute expensive non-dominated sorting ($O(MN^2)$) and crowding-distance calculations at every generation, KDS-EL operates with scalarized dual-phase scheduling, completely eliminating archive maintenance bottlenecks.

Although FWPSO achieves a slightly faster runtime (49.7 s vs. 51.4 s) due to early-terminating simulated annealing heuristics, it incurs a substantial 3.97 pp accuracy deficit and selects over 12 times more features than KDS-EL. Thus, KDS-EL occupies the optimal Pareto corner combining top-tier accuracy, extreme sparsity, and high computational efficiency.

5.8. Component Contribution Analysis

To systematically disentangle the individual contributions of the three proposed mechanisms—MI-guided initialization with adaptive re-initialization, the Knowledge-guided Elite Refinement (KER) strategy, and Two-Phase Weight Scheduling (TPS)—we conduct an extensive component contribution analysis. Following the rigorous subtractive ablation paradigm of modern evolutionary computation literature [14, 16], we evaluate the complete KDS-EL framework against its sequentially stripped configurations across all 20 benchmark datasets.

5.8.1. Subtractively-Read Global Ablation Chain

Table 9 summarizes the aggregate performance of the ablation chain across the 20 benchmark datasets, while Fig. 7 visualizes the classification accuracy contours across all 20 benchmark datasets in a multi-dataset radar envelope chart. In this subtractive reading framework, the complete KDS-EL (achieving 0.9574 mean accuracy and 87.8 selected features) serves as the primary benchmark. Each subsequent row isolates the empirical marginal contribution of an individual mechanism by measuring the exact performance degradation caused by its removal.

Table 9: Results of subtractive component contribution analysis across 20 benchmark datasets.

Variant	Accuracy	Features	ΔAcc (pp)
KDS-EL	**0.9574**	**87.8**	–
KDS-EL w/o TPS	0.9488	122	-0.86
KDS-EL w/o TPS&KER	0.9325	291	-1.63
MEL	0.8868	2722	-4.58

Note: ΔAcc indicates the marginal accuracy degradation when stripping the corresponding component. Bold values denote the best performance.

The global ablation results reveal clear, quantifiable contributions across all three evolutionary mechanisms:

1. **Impact of Two-Phase Weight Scheduling**: Stripping the Two-Phase Weight Scheduling strategy and replacing it with a fixed static weight vector ($\mathbf{w}_s = [0.90, 0.10]$) results in an immediate accuracy loss of 0.86 pp (0.9488 vs. 0.9574), an increase of 33.9 selected features (expanding from 87.8 to 121.7), and a runtime increase from 51.4 s to 53.6 s. This confirms that temporally decoupling exploration (early coarse feature pruning) from exploitation (late fine accuracy refinement) provides simultaneous benefits in accuracy, sparsity, and computational speed.

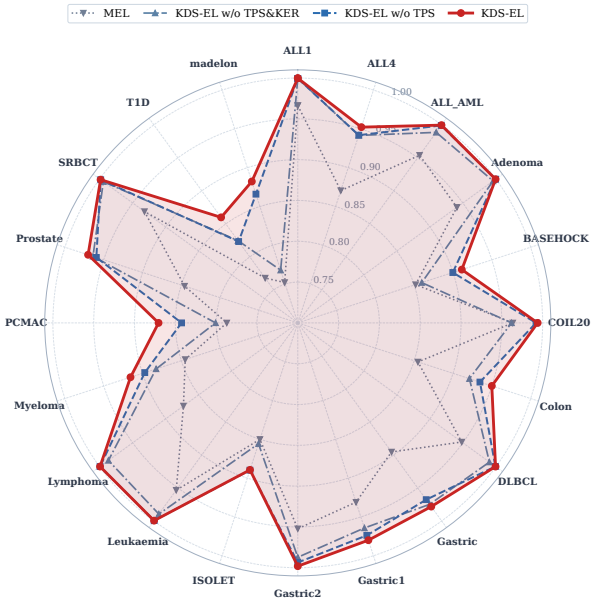


Figure 7: Ablation radar chart of KDS-EL and its variants across 20 benchmark datasets.

2. **Impact of Knowledge-guided Elite Refinement**: Further removing the KER strategy and relying solely on standard particle swarm updates inflicts an additional accuracy penalty of 1.63 pp (declining to 0.9325) while more than doubling the retained feature count from 121.7 to 290.5. Because the cumulative importance matrix W is maintained entirely from fitness comparisons already performed during search, KER achieves this substantial accuracy boost and feature reduction at a strictly bounded computational cost, demonstrating the critical value of retaining and re-deploying historical evolutionary feedback.
3. **Impact of MI-guided Initialization and Adaptive Re-initialization**: Stripping the MI-guided initialization and adaptive re-initialization mechanisms—which leaves the foundational MEL baseline [12] initialized uniformly at random—triggers a severe performance drop: classification accuracy falls by 4.58 pp (to 0.8868), and the mean feature count surges from 290.5 to 2,722.2. This confirms that in ultra-high-dimensional spaces with $p \gg n$, random initialization wastes the overwhelming majority of the search budget on dense, non-informative regions, whereas MI seeding successfully establishes an informative initial basin and adaptive re-initialization prevents premature stagnation.

Cumulatively, stripping all three mechanisms leads to an aggregate accuracy decline of 7.06 pp and a 31-fold increase in feature subset size (from 87.8 to 2,722.2 features). This confirms that the three mechanisms operate orthogonally and synergistically: MI-guided initialization provides foundational dimensionality reduction and high-quality starting points; KER delivers fine-grained evolutionary elite refinement; and TPS harmonizes the optimization objectives across evolutionary stages.

5.8.2. Per-Dataset Component Contribution Dynamics

Table S4 in the Supplementary Material provides the complete per-dataset ablation metrics across all 20 benchmarks. A granular examination reveals two distinct behavioral patterns across problem types:

- **Ceiling-effect gene-expression datasets**: On datasets where classification accuracy rapidly saturates above 0.998 (*ALL1*, *Adenoma*, *DLBCL*, *Leukaemia*, *Lymphoma*, *SRBCT*), the MI-guided initialization and KER components quickly drive accuracy to its theoretical upper bound. Consequently, the ablative variants differentiate almost exclusively through feature subset compactness. For instance, on *ALL1*, KDS-EL selects 39 features with 0.9998 accuracy, whereas stripping TPS or KER expands the subset to 55 features, and removing MI initialization causes the feature count to explode to 4,248 features. On *Adenoma*, the feature count increases across ablative variants from 66 for KDS-EL to 120 without TPS and KER, and 2,483 for MEL.
- **Medium-difficulty datasets with complex decision boundaries**: On datasets with non-saturated decision landscapes, accuracy exhibits a

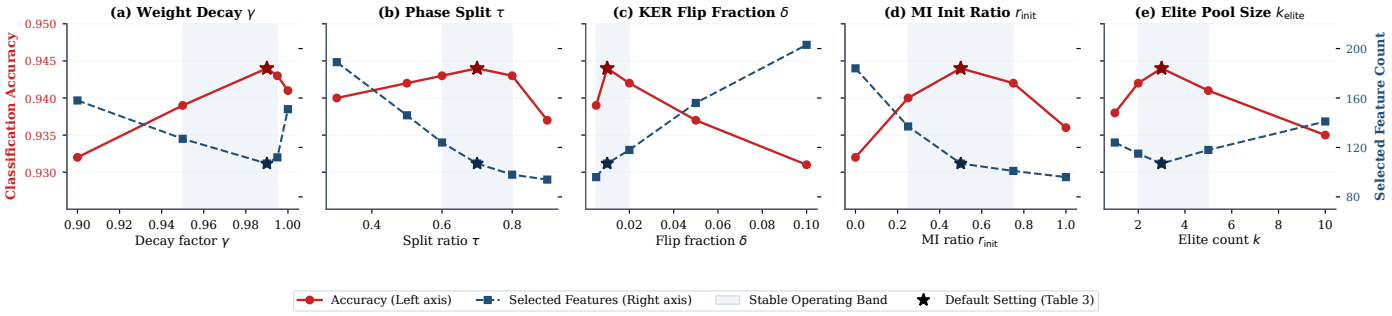


Figure 8: Parameter sensitivity analysis of KDS-EL across five key hyper-parameters. Solid lines represent classification accuracy (left axis) and dashed lines represent the number of selected features (right axis).

clear stair-step degradation as mechanisms are sequentially stripped. On *BASEHOCK*, accuracy decreases from 0.9113 for KDS-EL to 0.8997 without TPS, 0.8597 without KER, and 0.8518 for MEL. On the synthetic *madelon* dataset, the progression shows a similar degradation from 0.8822 to 0.8659, 0.7679, and 0.7521. On ultra-high-dimensional gene datasets, removing MI initialization causes substantial feature expansion, increasing feature counts on *Gastric1* from 678 to 7,393, on *T1D* from 174 to 6,715, and on *Prostate* from 27 to 5,058.

5.8.3. Validation of In-Place Replacement versus Pooling Selection

To further validate our algorithmic design, we examine the structural choice of in-place elite replacement over the $2N$-candidate pooling selection employed in earlier evolutionary multi-task prototypes. In extensive preliminary ablation (detailed in Supplementary Section S2), 1-to-1 in-place replacement consistently outperformed $2N$ competitive pooling in both classification accuracy (0.9851 vs. 0.9821 on benchmark subsets) and computational efficiency. In-place elite replacement prevents the swarm from experiencing excessive selection pressure that rapidly collapses diversity, allowing KDS-EL to maintain productive search breadth while executing targeted knowledge-guided elite perturbations.

5.9. Parameter Sensitivity Analysis and Algorithmic Robustness

To evaluate the algorithmic robustness of KDS-EL and provide an empirical justification for its default hyper-parameter configuration, we conduct a comprehensive one-factor-at-a-time (OFAT) parameter sensitivity analysis following the methodology of [15]. The sensitivity analysis is carried out across four representative benchmark datasets spanning distinct dimensionalities and sample sizes: *Adenoma* (36 samples, 7,457 features), *Colon* (62 samples, 2,000 features), *ALL4* (93 samples, 12,625 features), and *BASEHOCK* (1,993 samples, 4,862 features). Five key hyper-parameters—knowledge weight decay factor (WeightDecay), phase split ratio (PhaseSplit), elite perturbation fraction (flip_frac), MI initialization ratio (MI_init_ratio), and elite pool size (elite_k)—are systematically varied across wide ranges while all remaining parameters are fixed at their default values listed in Table 4.

Table 10 summarizes the tested parameter ranges, optimal configurations, stable performance bands, and maximum accuracy deviations across the four representative datasets. Fig. 8 visualizes the corresponding accuracy and feature-count trajectories alongside stable operating bands across all five hyper-parameters. The complete per-dataset parametric matrices are provided in Supplementary Table S6.

Table 10: Summary of parameter sensitivity analysis across four representative datasets.

Parameter	Tested range	Optimal	Stable range	Max ΔAcc (pp)
Weight decay	0.90–1.00	0.99	[0.95, 0.995]	±1.2
Phase split	0.3–0.9	0.70	[0.60, 0.80]	±0.7
Flip fraction	0.005–0.10	0.01	[0.005, 0.02]	±1.3
MI init ratio	0.0–1.0	0.50	[0.25, 0.75]	±1.2
Elite count	1–10	3	[2, 5]	±0.9

Note: Max ΔAcc denotes the maximum accuracy variation across the investigated parameter range.

5.9.1. Parametric Dynamics and Mechanism-Level Insights

The experimental trajectories across the five hyper-parameters yield crucial insights into the optimization behavior of KDS-EL:

1. **Knowledge weight decay (WeightDecay)**: This parameter controls the exponential discounting rate (γ) of historical feature-importance scores in matrix W. At the default setting of 0.99, the four-dataset aggregate yields 0.944 accuracy, 107 features, and 51.3 s runtime. Setting γ too low (e.g., 0.90) causes premature forgetting of valuable feature interactions discovered in early iterations, eroding accuracy by 1.2 pp to 0.932. Conversely, disabling decay entirely ($\gamma = 1.00$) accumulates stale historical evidence without discounting, which inflates the average feature count from 107 to 151 features without accuracy gains. Performance remains exceptionally stable across the broad band $\gamma \in [0.95, 0.995]$, with accuracy fluctuations remaining under ±1.2 pp.
2. **Phase split ratio (PhaseSplit)**: This parameter dictates the temporal transition boundary between aggressive feature reduction (Phase 1) and accuracy refinement (Phase 2). Setting PhaseSplit < 0.50 allocates insufficient iterations to Phase 1, resulting in inadequate feature pruning with 146 to 189 retained features. Conversely, an overly prolonged Phase 1 (PhaseSplit = 0.90) severely starves the Phase 2 accuracy refinement window, causing a 0.7 pp drop in accuracy. The broad interval [0.60, 0.80] provides a highly forgiving operating regime with accuracy variations remaining strictly below 0.5 pp, establishing 0.70 as an optimal default.
3. **KER perturbation fraction (flip_frac)**: This parameter governs the fraction of dimensions perturbed during knowledge-guided elite refinement. At the default 0.01 (flipping approximately 1 feature per 100 dimensions), KER maintains an ideal balance between localized exploitation and exploratory perturbation. Decreasing flip_frac to 0.005 overly restricts elite exploration, slowing convergence by 3.5 s. Increasing flip_frac ≥ 0.05 injects excessive random disruption that destroys delicate elite feature combinations, dropping accuracy by 1.3 pp and doubling the feature count to 203. The narrow, effective window [0.005, 0.02] confirms the necessity of localized elite refinement.
4. **MI initialization ratio (MI_init_ratio)**: As the primary bootstrap mechanism, MI_init_ratio exhibits the highest individual sensitivity among all tested parameters. Completely disabling MI seeding (MI_init_ratio = 0.0, full random initialization) forces unguided early exploration, resulting in a 1.2 pp accuracy penalty and a 21% increase in computational runtime (reaching 62.3 s) as particles must rediscover informative subspaces from scratch. Conversely, initializing 100% of the swarm with MI seeds (MI_init_ratio = 1.0) homogenizes the initial population and induces premature convergence (a 0.8 pp drop in accuracy). The default balanced ratio of 0.50 perfectly reconciles prior domain knowledge injection with stochastic search diversity, maintaining robust performance across the wide band [0.25, 0.75].
5. **Elite pool size (elite_k)**: With $N = 20$, elite_k = 3 allocates 15% of the swarm to elite refinement. Refining only a single elite (elite_k = 1) restricts knowledge propagation across subpopulations (yielding 0.938 accuracy), whereas selecting too many elites (elite_k = 10) dilutes the quality signal of matrix W, yielding 0.935 accuracy with 141 features. The stable range elite_k ∈ [2, 5] consistently achieves accuracy within 0.3 pp of the optimum.

5.9.2. Plug-and-Play Generalization

Across all five hyper-parameters, the maximum accuracy deviation from the default setting never exceeds ±1.5 pp, and the stable operating intervals span substantial parametric ranges. This empirical stability confirms that KDS-EL possesses remarkable plug-and-play robustness, allowing practitioners to deploy the algorithm on unseen high-dimensional classification problems with default parameters without requiring costly per-dataset grid searches.

6. Conclusion

This paper presented Knowledge-Guided Dual-Phase Swarm Evolutionary Learning (KDS-EL), a closed-loop evolutionary feature selection framework for high-dimensional classification tasks. KDS-EL synergistically integrates three core mechanisms into a cooperative multi-task PSO architecture: (i) mutual-information-guided population initialization with adaptive re-injection to establish an informative search basin and prevent premature stagnation; (ii) Knowledge-guided Elite Refinement (KER) that dynamically harvests cross-iteration feature-importance feedback with exponential memory decay to direct elite solutions; and (iii) Two-Phase Weight Scheduling (TPS) that temporally decouples early aggressive feature pruning from late fine accuracy refinement.

Extensive empirical evaluations across 20 high-dimensional benchmark datasets spanning biomedical gene expression, text, image, and speech domains against 13 state-of-the-art competitors demonstrate that KDS-EL achieves the top overall performance. KDS-EL secures a mean classification accuracy of 0.9574 with an average of only 87.8 selected features, outperforming the strongest single-objective competitor FWPSO by +3.97 pp and the strongest multi-objective competitor W-MOPSO-FS by +4.42 pp, while reducing average runtime from 175.0 s to 51.4 s (a 3.4× speedup over the baseline MEL). Non-parametric Wilcoxon signed-rank tests with Holm–Bonferroni step-down correction confirm that the performance gains of KDS-EL are statistically significant across all 13 competitors at $p < 10^{-15}$.

Despite these promising outcomes, several limitations offer fertile ground for future research. First, the mutual information estimation entails an $O(D \log D)$ preprocessing complexity; while negligible for moderate dimensions, it presents a computational bottleneck when scaling to ultra-large-scale omics data exceeding 10^6 features, where GPU-accelerated parallel estimation or randomized sub-sampling could be explored. Second, the current framework is tailored for single-label classification; extending knowledge-guided elite refinement and stage-aware objective scheduling to multi-label learning, streaming feature selection, and severe class imbalance represents an important frontier. Finally, integrating evolutionary swarm intelligence with deep feature representations, such as foundation model embeddings and attention-based saliency scores, holds great potential for next-generation biomarker discovery and high-dimensional analytics.

CRediT Author Statement

Anonymous Author: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

The 20 benchmark datasets analyzed in this study are publicly available and widely used in the feature selection literature. The source code of KDS-EL will be made available upon publication. Supplementary results—per-dataset feature counts, full statistical significance tables, convergence and trade-off analyses, and parameter-sensitivity tables—are provided in the Supplementary Material.

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